

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 27/11

FULL LENGTH TEST

Solution

English



BPSC TEST-27/11 (SOLUTION)

Q 1. Ans. C

Explanation:

- **Beta (β) decay** is a type of radioactive decay in which a nucleus emits a **beta particle** (an electron or a positron) along with a neutrino or antineutrino. This process occurs due to the **weak nuclear force**, one of the four fundamental forces of nature. The **strong nuclear force** binds protons and neutrons inside the nucleus, the **electromagnetic force** acts between charged particles, and the **gravitational force** is negligible at the nuclear scale. Therefore, beta decay is governed by the **weak nuclear force**.

Q 2. Ans. C

Explanation:

- **Streamline flow** (also called **laminar flow**) is a type of fluid motion in which **every fluid particle follows a smooth, well-defined path**, and the **velocity of the fluid at any given point remains constant with time**. Although the velocity may differ from one point to another, it does not change with time at a particular point. This makes the flow orderly and predictable.
- In contrast, when the fluid forms **eddies and whirlpools** and the velocity changes randomly with time, the flow is called **turbulent flow**. Therefore, streamline flow is characterised by **constant velocity at each point over time**, not by random fluctuations. The direction of flow is along the streamlines, not perpendicular to them.

Q 3. Ans. A

Explanation:

- **Statement 1 is correct:** **Surface tension** is the property of a liquid surface that makes it behave like a stretched elastic membrane. A liquid always tends to **minimise its surface area** because this corresponds to the minimum surface energy. Among all shapes with the same volume, a **sphere has the smallest surface area**. Therefore, small liquid drops such as raindrops and mercury droplets naturally assume a **spherical shape** due to surface tension.
- **Statement 2 is incorrect:** the **surface tension of a liquid decreases as temperature increases**. When a liquid is heated, the kinetic energy of its molecules

increases, weakening the intermolecular cohesive forces responsible for surface tension. As a result, the liquid surface becomes less tightly held together, causing the surface tension to decrease. Near the critical temperature, the surface tension approaches zero. Therefore, this statement is incorrect.

Q 4. Ans. B

Explanation:

- **Young's modulus** measures the elasticity of a material when it is subjected to **tensile or compressive forces** acting along its length. It is defined as the **ratio of longitudinal (normal) stress to longitudinal strain**, provided the elastic limit of the material is not exceeded.
- Young's modulus describes how resistant a material is to changes in its length. A material with a **high Young's modulus** (such as steel) undergoes only a small deformation under a given force, whereas a material with a **low Young's modulus** (such as rubber) deforms much more easily.

Q 5. Ans. D

Explanation:

- The **energy of an individual photon of electromagnetic radiation is determined by the frequency of its photons**. The energy of a single photon is given by Planck's equation:
✓ $E = hf$ Where:
 - E = Energy of the photon
 - h = Planck's constant
 - f = Frequency of the electromagnetic wave
- As the **frequency increases**, the energy of each photon also increases. Conversely, electromagnetic waves with lower frequencies have lower photon energies. For example, **gamma rays** have the highest frequency and therefore the highest energy, whereas **radio waves** have the lowest frequency and the lowest energy.
- Although **amplitude** determines the **intensity (brightness)** of an electromagnetic wave, it does not determine the energy of an individual photon. The **speed** of electromagnetic waves in a vacuum is constant (3×10^8 m/s), and the **wavelength** is inversely proportional to frequency. Thus, it is the **frequency** that directly determines the energy of an electromagnetic wave.

Q 6. Ans. B**Explanation:**

- **Sound intensity** is defined as the **sound power transmitted per unit area** perpendicular to the direction of wave propagation. Its SI unit is **watt per square metre (W/m^2)**. It represents the amount of sound energy passing through a unit area in one second. A higher sound intensity corresponds to a greater flow of sound energy.
- **Decibel (dB)** is **not a unit of sound intensity**; it is a **logarithmic unit used to express sound intensity level or loudness** relative to a reference intensity.
- **Newton/ m^2 (N/m^2)** is the SI unit of **pressure (Pascal)**, not sound intensity.
- **Joule/ m^2 (J/m^2)** is the unit of **energy per unit area or surface energy**, not sound intensity.

Q 7. Ans. D**Explanation:**

- **Pair 1 is correct:** **alpha (α) particles** are helium nuclei consisting of two protons and two neutrons. Because they are relatively heavy and carry a +2 charge, they interact strongly with matter and lose energy rapidly. As a result, they have the **least penetrating power** among the three types of radioactive radiations. Alpha particles can be stopped by a sheet of paper or even the outer layer of human skin.
- **Pair 2 is correct:** **Beta (β) rays** are high-speed electrons or positrons emitted from the nucleus during radioactive decay. They have greater penetrating power than alpha particles but less than gamma rays. Beta particles can pass through paper but are effectively stopped by a few millimetres of aluminium. Hence, beta rays possess **intermediate penetrating power**.
- **Pair 3 is correct:** **Gamma (γ) rays** are high-energy electromagnetic radiations emitted from an excited nucleus. They have no mass and no electric charge, allowing them to penetrate matter much more deeply than alpha and beta radiations. Thick layers of lead or several centimetres of concrete are required to absorb gamma rays effectively. Therefore, gamma rays have the **highest penetrating power**.
- **Pair 4 is correct:** **radioactivity** is the **spontaneous disintegration of unstable atomic nuclei** accompanied by the emission of alpha particles, beta particles, and/or gamma rays. This process occurs naturally without any external influence and leads to the formation of a more stable nucleus. Therefore, radioactivity is correctly described as **spontaneous nuclear decay**.

Q 8. Ans. A**Explanation:**

- **Statement 1 is correct:** **ethanoic acid (CH_3COOH)** belongs to the family of **carboxylic acids**. Its characteristic functional group is the **carboxyl group ($-\text{COOH}$)**, which consists of a carbonyl group ($\text{C}=\text{O}$) and a hydroxyl group ($-\text{OH}$) attached to the same carbon atom. The presence of this group gives ethanoic acid its acidic properties, such as turning blue litmus red and reacting with carbonates and hydrogen carbonates.
- **Statement 2 is correct:** **ethanoic acid reacts with sodium hydrogen carbonate (NaHCO_3)** to produce **carbon dioxide gas**, water, and sodium ethanoate. The evolution of carbon dioxide causes **brisk effervescence**, making this a simple test to identify carboxylic acids.
- **Statement 3 is incorrect:** **Ethanol is not acidic enough** to react with sodium hydrogen carbonate. Hence, it does **not** liberate carbon dioxide and **does not produce brisk effervescence**. Only acids such as ethanoic acid react with sodium hydrogen carbonate to release CO_2 gas. Therefore, this statement is incorrect.

Q 9. Ans. A**Explanation:**

- **Statement 1 is correct:** **hydrogenation** is the process of adding hydrogen to **unsaturated vegetable oils** in the presence of a suitable catalyst. During this process, the carbon-carbon double bonds in the oil are converted into single bonds, producing **saturated fats**. This reaction is widely used in the manufacture of **vanaspati ghee** and margarine. Therefore, this statement is correct.
- **Statement 2 is correct:** **finely divided nickel (Ni)** is commonly used as the catalyst in the hydrogenation of vegetable oils. The catalyst speeds up the addition of hydrogen to the unsaturated hydrocarbons without itself undergoing any permanent chemical change. Although catalysts such as platinum or palladium can also be used, **nickel is the standard catalyst**.
- **Statement 3 is incorrect:** Hydrogenation is **not an oxidation reaction**. It is a **reduction reaction** because it involves the **addition of hydrogen** to an unsaturated compound. In organic chemistry, the **addition of hydrogen is reduction**, while the **addition of oxygen or the removal of hydrogen is oxidation**. Therefore, hydrogenation is classified as a reduction process, making this statement incorrect.

Q 10. Ans. A

Explanation:

- **Assertion (A) is true:** a liquid boils when its vapour pressure becomes equal to the external pressure. Liquids with **stronger intermolecular forces** (such as hydrogen bonding or strong dipole-dipole attractions) hold their molecules together more tightly. Consequently, a **higher temperature** is required to provide sufficient energy for the molecules to overcome these attractive forces and escape into the vapour phase. Therefore, liquids with stronger intermolecular forces have **higher boiling points**.
- **Reason (R) is true:** Strong intermolecular forces increase the attraction between neighbouring molecules, so **more thermal energy** must be supplied to overcome these forces during the liquid-to-gas phase transition. Hence, the boiling point rises as the strength of intermolecular forces increases.
- **Since both the Assertion and the Reason are true, and the Reason correctly explains the Assertion, the correct answer is: Both (A) and (R) are true, and (R) is the correct explanation of (A).**

Q 11. Ans. C

Explanation:

- **Assertion (A) is true:** **Amorphous solids** lack a regular geometrical shape because their constituent particles are **not arranged in a long-range ordered pattern**. Unlike crystalline solids, they do not possess well-defined plane surfaces or characteristic geometrical forms. Materials such as **glass, rubber, and plastics** are common examples of amorphous solids. Therefore, the assertion is true.
- **Reason (R) is false:** **crystalline solids**, not amorphous solids. In crystalline solids, atoms, ions, or molecules are arranged in a **regular, repeating three-dimensional lattice**, giving them a definite geometrical shape and sharp melting point. In contrast, **amorphous solids have only short-range order**, with particles arranged randomly and irregularly. Therefore, the reason is false.

Q 12. Ans. A

Explanation:

- **Statement 1 is correct:** **Osmosis** is the movement of solvent molecules (usually water) through a semipermeable membrane from a region of lower solute concentration (higher solvent concentration) to a region of higher solute

concentration (lower solvent concentration) until equilibrium is reached. Therefore, the statement is correct.

- **Statement 2 is incorrect:** an **emulsion** is a colloidal system in which both the dispersed phase and the dispersion medium are liquids (liquid in liquid). Common examples include **milk (fat droplets dispersed in water)** and **cold cream (water dispersed in oil)**. A gas dispersed in a solid is not an emulsion; examples include **pumice stone** and **foam rubber**, which are classified as **solid foams**. Therefore, this statement is incorrect.

Q 13. Ans. B

Explanation:

- **Pair 1 is incorrect:** zinc reacts with dilute hydrochloric acid to produce **zinc chloride** and **hydrogen gas**, not carbon dioxide. Metals above hydrogen in the reactivity series displace hydrogen from dilute acids. The reaction is: $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$. Therefore, this pair is incorrectly matched because the gas evolved is **hydrogen**, not carbon dioxide.
- **Pair 2 is incorrect:** metal carbonates react with acids to form a **salt, water, and carbon dioxide**. When sodium carbonate reacts with dilute hydrochloric acid, **carbon dioxide** is liberated, not hydrogen. $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$ Therefore, this pair is incorrectly matched.
- **Pair 3 is correct:** Hydrochloric acid reacts with sodium hydroxide in a **neutralisation reaction**. During this reaction, the hydrogen ions (H^+) from the acid combine with hydroxide ions (OH^-) from the base to form **water**, while the remaining ions form **sodium chloride (NaCl)**. This is a typical acid-base reaction. $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. Hence, this pair is correctly matched.
- **Pair 4 is correct:** When **carbon dioxide** is passed through **lime water (calcium hydroxide solution)**, it reacts to form **insoluble calcium carbonate (CaCO_3)**, which appears as a **white precipitate** and turns the solution milky. $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3\downarrow + \text{H}_2\text{O}$. If excess carbon dioxide is passed, the white precipitate dissolves due to the formation of **soluble calcium bicarbonate**. Therefore, this pair is correctly matched.

Q 14. Ans. A

Explanation:

- **Euglena** exhibits a **dual (mixotrophic) mode of nutrition**, meaning it can obtain food by both

autotrophic and **heterotrophic** methods. It contains **chloroplasts** with chlorophyll and performs **photosynthesis** in the presence of sunlight, behaving like a plant. In the absence of light, however, it cannot photosynthesise and instead absorbs organic nutrients from its surroundings, behaving like an animal. This ability to switch between autotrophic and heterotrophic nutrition is known as **mixotrophic nutrition**.

Q 15. Ans. B

Explanation:

- **Haemophilia** is a **sex-linked recessive disorder** caused by a mutation in a gene located on the **X chromosome**. The disorder impairs the normal blood-clotting mechanism because of the deficiency of specific clotting factors. Since males possess only one X chromosome (XY), they develop the disease if they inherit the defective gene. Females (XX) usually become **carriers** because they must inherit two defective copies of the gene to express the disease. Thus, haemophilia follows an **X-linked recessive pattern of inheritance**.
- **Albinism** is generally an **autosomal recessive disorder** affecting melanin production and is not sex-linked.
- **Down syndrome** is a **chromosomal disorder** caused by the presence of an extra copy of chromosome 21 (**trisomy 21**), not by a recessive gene on the sex chromosomes.
- **Sickle cell anaemia** is an **autosomal recessive genetic disorder** caused by a mutation in the β -globin gene on chromosome 11 and is inherited equally by males and females.

Q 16. Ans. A

Explanation:

- **Assertion (A) is true:** **Brownian motion** is the continuous zigzag movement of colloidal particles caused by their unequal collisions with the molecules of the dispersion medium. Although molecules in a true solution also undergo random motion, they are too small to be observed individually. In contrast, **colloidal particles are large enough to be observed under an ultramicroscope**, making Brownian motion much more evident in colloidal systems.
- **Reason (R) is true:** Colloidal particles are **large enough to receive countless unequal collisions from the rapidly moving molecules of the dispersion medium**, yet **small enough to remain suspended** instead of settling under gravity. These continuous

unequal impacts produce the characteristic zigzag movement known as Brownian motion. Thus, the reason correctly explains why Brownian motion is observed more clearly in colloids than in true solutions.

- **Therefore, both the Assertion and the Reason are true, and the Reason is the correct explanation of the Assertion. Correct Answer: Both (A) and (R) are true, and (R) is the correct explanation of (A).**

Q 17. Ans. D

Explanation:

- **Auxins** are plant growth hormones primarily synthesized in the **shoot apex**. They promote **apical dominance**, a phenomenon in which the apical bud suppresses the growth of lateral buds, allowing the main shoot to grow more vigorously.
- **Cytokinins** stimulate **cell division** and are known to **delay leaf senescence** by promoting nutrient mobilization and maintaining chlorophyll content. This property helps keep leaves green for a longer duration.
- **Gibberellins** promote **elongation of stems and internodes**, especially in dwarf plants. They also stimulate seed germination, flowering in some plants, and fruit development.
- **Abscissic acid (ABA)** is a growth-inhibiting plant hormone. It induces **closure of stomata during water stress**, thereby reducing water loss through transpiration. It also promotes seed dormancy and helps plants tolerate adverse environmental conditions.

Q 18. Ans. C

Explanation:

- **B-lymphocytes (B cells)** are responsible for **humoral immunity**. After activation by an antigen (with the help of helper T cells), they differentiate into **plasma cells**, which produce antibodies. Thus, B-lymphocytes are associated with antibody production.
- **T-helper ($CD4^+$) cells** coordinate the immune response by releasing cytokines that activate B cells, cytotoxic T cells, and macrophages. They are part of the **cell-mediated immune response**, which is carried out by T lymphocytes.
- Macrophages are **antigen-presenting cells (APCs)**. They engulf pathogens by phagocytosis, process the antigens, and present them on **MHC Class II molecules** to helper T cells, thereby initiating the adaptive immune response.

- Plasma cells are the **activated, differentiated form of B-lymphocytes**. Their primary function is the **secretion of large quantities of antibodies**, making them the body's main antibody-producing cells.

Q 19. Ans. C

Explanation:

- **Elephantiasis**, also known as **lymphatic filariasis**, is a parasitic disease caused mainly by the filarial worms **Wuchereria bancrofti** and **Brugia malayi**. These worms live in the **lymphatic vessels** of infected individuals, causing blockage of lymph flow and resulting in severe swelling of the limbs and genital organs.
- The disease is **transmitted to humans through the bite of an infected female Culex mosquito**, which acts as the **vector**. During a blood meal, the mosquito introduces the infective larval stage of the parasite into the human body. The larvae then migrate to the lymphatic system, where they mature into adult worms and cause the characteristic symptoms of elephantiasis.

Q 20. Ans. B

Explanation:

- **Vitamin B₃ (Niacin)** is an essential water-soluble vitamin that plays a key role in energy metabolism as a component of the coenzymes **NAD (Nicotinamide Adenine Dinucleotide)** and **NADP (Nicotinamide Adenine Dinucleotide Phosphate)**. These coenzymes participate in numerous oxidation-reduction reactions necessary for the normal functioning of cells.
- A deficiency of niacin causes **pellagra**, a nutritional deficiency disease characterised by the classical "three Ds":
 - ✓ **Dermatitis** (inflamed, scaly skin, especially on sun-exposed areas),
 - ✓ **Diarrhoea**, and
 - ✓ **Dementia**.
- If left untreated, pellagra can eventually lead to death. The disease is commonly associated with diets deficient in niacin or the amino acid **tryptophan**, from which the body can synthesise niacin.

Q 21. Ans. D

Explanation:

- **Implantation** is the process by which the **blastocyst (early embryo)** attaches to and becomes embedded in the **endometrium**, the inner lining of the **uterus**. After fertilisation occurs in the **ampullary-isthmic**

junction of the fallopian tube, the developing embryo undergoes repeated cell divisions while moving toward the uterus. Around **6–7 days after fertilisation**, the blastocyst reaches the uterus and implants into the thick, vascular endometrium.

- Implantation is essential for the establishment of pregnancy because it enables the developing embryo to receive nourishment and oxygen from the mother through the subsequent formation of the **placenta**.

Q 22. Ans. C

Explanation:

- **Maize** is a **monocotyledonous plant**, and monocots characteristically possess a **fibrous root system**. In this type of root system, the primary root is short-lived and is replaced by a large number of thin, similarly sized adventitious roots arising from the base of the stem. These roots spread widely in the soil and provide efficient absorption of water and minerals as well as support to the plant.
- The other options are **dicotyledonous plants**, which possess a **tap root system**:
 - ✓ **Rose** has a tap root.
 - ✓ **Mango** has a well-developed tap root.
 - ✓ **Mustard** also possesses a tap root system.
- Thus, among the given options, only **maize** has a **fibrous root system**.

Q 23. Ans. D

Explanation:

- **Statement 1 is incorrect:** fungi **do not contain chlorophyll** and are therefore **unable to perform photosynthesis**. They are **heterotrophic organisms** that obtain nutrients by absorbing dissolved organic matter from dead organisms (saprophytes), living hosts (parasites), or through mutualistic associations such as lichens and mycorrhizae. Hence, this statement is incorrect.
- **Statement 2 is incorrect:** The **cell wall of fungi is mainly composed of chitin and polysaccharides**, not cellulose. **Cellulose** is the principal structural component of the cell walls of **plants**, whereas **chitin** provides rigidity and strength to fungal cell walls. Therefore, this statement is incorrect.
- **Statement 3 is correct:** fungi store reserve food materials in the form of **glycogen and oils**, unlike plants, which primarily store **starch**. The storage of glycogen is one of the characteristics that fungi share with animals, reflecting their distinct mode of nutrition and metabolism.

Q 24. Ans. D

Explanation:

- **Pair 1 is correct:** **sericulture** is the practice of rearing silkworms for the commercial production of silk. It is an important agro-based industry that involves the cultivation of mulberry plants and the breeding of silkworms, mainly *Bombyx mori*, for obtaining silk fibres.
- **Pair 2 is correct:** **horticulture** is the branch of agriculture concerned with the cultivation of fruits, vegetables, flowers, and ornamental plants. It includes several specialized branches such as pomology (fruits), olericulture (vegetables), floriculture (flowers), and landscape gardening. Therefore, associating horticulture with fruits and vegetables is correct.
- **Pair 3 is correct:** **Pomology** is a specialized branch of horticulture that deals with the cultivation, production, improvement, harvesting, storage, and study of fruit crops. It focuses on fruit-bearing plants such as mango, apple, banana, citrus, and grapes. Hence, this pair is correctly matched.
- **Pair 4 is correct:** **Olericulture** is the branch of horticulture concerned with the cultivation and production of vegetable crops, particularly non-woody (herbaceous) vegetables such as cabbage, spinach, tomato, brinjal, cauliflower, and carrot. Therefore, this pair is also correctly matched.

Q 25. Ans. B

Explanation:

- **Statement:** xylem vessels are dead cells at maturity with thick, lignified walls that conduct water and minerals from roots to aerial parts of the plant. In contrast, phloem sieve tube elements are living cells (although they lack a nucleus at maturity) and are assisted by companion cells. They transport organic food materials such as sucrose throughout the plant.
- **Conclusion I:** xylem vessels are dead at maturity and yet efficiently conduct water and minerals. Thus, transport in plants is not restricted to living tissues alone. Both living (phloem) and dead (xylem) conducting tissues participate in transport, depending on the material being transported.
- **Conclusion II:** Water and mineral transport occurs through dead xylem elements and is driven mainly by physical forces such as transpiration pull. In contrast, food transport through phloem requires living sieve tube elements and companion

cells because translocation is an energy-dependent (active) process. Hence, the requirement for living cells depends on the nature of the substance being transported.

Q 26. Ans. B

Explanation:

- **Varya is India's first indigenous distilled video AI model**, developed by Avataar with support from the IndiaAI Mission. It converts text and images into high-quality videos at a significantly lower cost, making generative AI more accessible and better suited to Indian contexts such as regional attire, festivals, and architecture. Using a distillation technique, Varya reduces the video generation process from 50 steps to just 4, enabling video creation at approximately ₹0.48 per second. The model is designed to provide affordable, localized visual content for educators, MSMEs, and e-commerce businesses.

Q 27. Ans. C

Explanation:

- **Statement:** mature human red blood cells (RBCs) are enucleated, meaning they lose their nucleus during maturation. This adaptation enables them to transport oxygen more efficiently and contributes to their characteristic biconcave shape.
- **Conclusion I:** Since mitosis requires a nucleus containing DNA, mature RBCs are incapable of cell division. Once they lose their nucleus during development, they can no longer replicate or synthesize new proteins. Their lifespan is about 120 days, after which they are replaced by new RBCs formed in the red bone marrow.
- **Conclusion II:** The absence of a nucleus creates additional intracellular space, allowing mature RBCs to accommodate a larger quantity of haemoglobin, the oxygen-carrying pigment. This adaptation significantly increases the oxygen transport capacity of each red blood cell.

Q 28. Ans. A

Explanation:

- **Statement 1 is correct:** Aditya-L1 is India's first space-based solar mission to study the Sun. It was launched on 2 September 2023 by the Indian Space Research Organisation (ISRO) from the Satish Dhawan Space Centre, Sriharikota.
- **Statement 2 is correct:** The spacecraft has been placed in a halo orbit around the Sun–Earth L1

(Lagrange Point-1). The L1 point, located about 1.5 million km from Earth towards the Sun, provides an uninterrupted view of the Sun without frequent eclipses or occultation, making it an ideal location for continuous solar observations.

- **Statement 3 is incorrect:** The mission was **not** launched using GSLV Mk III (LVM3). Instead, Aditya-L1 was launched aboard the PSLV-XL (Polar Satellite Launch Vehicle-XL). The spacecraft carries seven scientific payloads, including visible and X-ray spectrometers, a coronagraph, solar wind and energetic particle analysers, and magnetometers to study the Sun's corona, photosphere, chromosphere, solar wind, and magnetic field.

Q 29. Ans. B

Explanation:

- The Fram2 mission was a landmark **private SpaceX** spaceflight that became the **first mission to send an all-civilian crew into polar orbit**. Funded by entrepreneur Chun Wang, the four-member crew launched aboard a **SpaceX Crew Dragon** spacecraft on **31 March 2025** from NASA's **Kennedy Space Center, Florida**. During their nearly **four-day mission**, the crew completed **22 scientific experiments**, including capturing the **first X-ray image in space**, while travelling in a **90-degree polar orbit** over both the **North and South Poles**. The unique trajectory enabled observations of rare atmospheric phenomena such as **STEVE (Strong Thermal Emission Velocity Enhancement)**. The mission concluded successfully with a **splashdown** on **4 April 2025** off the coast of **Oceanside, California**, marking the **first Crew Dragon recovery in the Pacific Ocean**.

Q 30. Ans. A

Explanation:

- **Statement:** DNA replication takes place during the **S phase of interphase**, which occurs **before meiosis I begins**. As a result, each chromosome consists of **two identical sister chromatids** at the start of meiosis.
- **Conclusion I is incorrect:** While chromosomes enter meiosis in a replicated state, the second meiotic division (**Meiosis II**) separates the sister chromatids. As a result, the **final four** daughter cells receive **unreplicated** chromosomes (single chromatids), not replicated ones.
- **Conclusion II is incorrect:** No DNA replication

occurs **between meiosis I and meiosis II**. The interval between these two divisions is called **interkinesis**, during which DNA synthesis does **not** take place. Meiosis II follows directly after meiosis I without another round of DNA replication.

Q 31. Ans. A

Explanation:

- **Statement 1 is correct:** A major continuing highlight of the PLFS data is the sustained increase in female workforce participation. The Female LFPR (usual status, 15 years and above) has been rising steadily over the last few years (reaching 41.7% in 2023–24) and maintained its position above the 40% threshold in the 2025 report, driven largely by increased rural female participation.
- **Statement 2 is incorrect:** The Worker Population Ratio (WPR), which indicates the percentage of employed persons in the population, did **not decline**. The PLFS 2025 shows a continued increasing trend in the overall WPR for persons aged 15 years and above, reflecting steady labour absorption in the economy.
- **Statement 3 is incorrect:** Structurally, in the Indian economy, the urban unemployment rate is consistently **higher** than the rural unemployment rate. According to the PLFS 2025 findings, while the overall UR hovered around 3.1%, the rural UR was significantly lower (around 2.4%) compared to the urban UR (around 4.8%), largely because rural employment is heavily supported by agriculture and informal labour.

Q 32. Ans. B

Explanation:

- **Statement A is correct:** The statement accurately identifies the proposed functionality of the new service.
 - ✓ The RBI is exploring the feasibility of a centralised 'Payments Switching Service' (PaSS). This is designed to solve the challenges customers face when their account numbers change due to shifting banks or bank mergers.
 - ✓ It facilitates the frictionless migration of both incoming and outgoing payment instructions from one account to another.
 - ✓ Customers will be able to view all their linked payment flows and initiate either a full or partial switch with appropriate authorisation.
 - ✓ The ultimate objectives are to provide customers with centralised control, ensure

orderly transitions during systemic changes (like mergers), and foster healthy competition and service excellence among financial institutions.

- **Statement B is incorrect:** The statement falsely claims the beneficiary's bank will exclusively bear the liability.

- ✓ Currently, the existing Limiting Liability Framework places the responsibility and liability exclusively on the customer's bank (the issuer).
- ✓ To ensure balanced accountability, the RBI is introducing a Shared Responsibility Framework. Under this new framework, both the customer's bank (issuer) and the beneficiary's bank will jointly bear the liability for unauthorised digital payment transactions.
- ✓ The objective behind this shift is to financially incentivise both parties to implement robust fraud detection and prevention measures and to strengthen inter-bank coordination for timely interventions, thereby boosting consumer trust.

- **Statement C is correct:** The statement accurately reflects the RBI's plan to consolidate the authorisation process for international transactions.

- ✓ **Currently, entities seeking to facilitate end-to-end cross-border funds transfers must seek dual authorisations:** one under the Payment and Settlement Systems (PSS) Act, 2007, and an authorised person license under the Foreign Exchange Management Act (FEMA), 1999.
- ✓ To gain efficiencies, the RBI is examining a single-window application process.
- ✓ This streamlining is intended to significantly enhance the ease of doing business for entities in the cross-border ecosystem and promote newer use cases and technological innovation.

- **Statement D is correct:** The statement correctly identifies the regulatory exemption proposed for a new tier of payment operators.

- ✓ Recognising that the payments space is highly dynamic, the RBI is exploring the creation of a new class called Small Payment System Providers (SPSPs).
- ✓ Under a perpetual regulatory sandbox structure, these innovative solutions would not require prior authorisation by the RBI to begin carrying out their activities.
- ✓ Instead of being regulated from the start, they will only be brought into the regulatory fold if

and when their activities reach a specified level of volume, significance, or criticality.

- ✓ This represents a shift toward a tailored, risk-calibrated oversight mechanism designed to promote the ease of doing business without sacrificing systemic stability.

Q 33. Ans. B

Explanation:

- **Pair 1 is correctly matched:** In the curated Vishwa Sutra collection, Muga silk, a traditional, naturally golden silk renowned for its durability and glossy texture from Assam, was creatively reimaged and paired with Egyptian design elements and cultural silhouettes to appeal to the global fashion market.
- **Pair 2 is correctly matched:** The iconic Kanchipuram weave from Tamil Nadu, traditionally characterised by its heavy silk and intricate zari work, was styled and reinterpreted for this initiative by integrating minimalist Norwegian lines and aesthetic influences.
- **Pair 3 is incorrectly matched:** Odisha Ikat was actually paired with Greek forms and drapes, whereas it was the double-ikat Patola weave from Gujarat that was reimaged globally using the Spanish design influences.
- **Pair 4 is incorrectly matched:** In the Vishwa Sutra initiative, the rich Banarasi weave was reimaged and paired specifically with UAE-inspired ensembles, while the Greek cultural silhouettes were combined with the Odisha Ikat as established above.

Q 34. Ans. C

Explanation:

- General Anti-Avoidance Rules (GAAR) are provisions under the Income Tax Act that empower revenue authorities to deny tax benefits to arrangements or transactions whose primary purpose is to avoid taxes and which lack bona fide commercial substance.
- The Government of India recently amended the Income Tax Rules, 2026 (under the Income Tax Act, 2025) to introduce a "grandfathering" clause. This amendment explicitly excludes investments made prior to **1 April 2017** from the purview of GAAR, thereby protecting older, legitimate investments from retrospective tax scrutiny.
- **Key Committee:** The implementation and framework of GAAR in India were heavily shaped by the recommendations of the **Parthasarathi Shome Committee (2012)**.

- It is primarily used to combat aggressive tax planning, “treaty shopping” (routing investments through low-tax jurisdictions like Mauritius or the Cayman Islands purely to avoid domestic taxes), and round-tripping of funds.
- To prevent the harassment of small taxpayers, GAAR provisions are generally invoked only when the tax benefit from the arrangement crosses a specific threshold (currently ₹3 crore).
- GAAR aligns India with the OECD’s **Base Erosion and Profit Shifting (BEPS)** framework, which ensures that profits are taxed where the actual economic activities and value creation occur.

Q 35. Ans. A

Explanation:

- **Statement 1 is correct:** The Urban Challenge Fund is structured to operate within a definitive timeline, providing a total financial allocation of ₹1 lakh crore in Central Assistance spanning from financial year 2025–26 to 2030–31.
- **Statement 2 is correct:** To enforce financial independence and market discipline, the fund mandates that beneficiary cities must source a minimum of 50% of their total project costs through market mechanisms like municipal bonds, bank loans, or Public-Private Partnerships.
- **Statement 3 is incorrect:** The framework is designed to leverage finance rather than fully fund it, meaning that Central Assistance will cover only 25% of the total project cost instead of a 75% share.
- **Statement 4 is incorrect:** While a dedicated Credit Repayment Guarantee Scheme has been approved to assist smaller and vulnerable urban local bodies with creditworthiness, its specific allocated corpus is ₹5,000 crore, not ₹50,000 crore.
- **Additional Facts:**
 - ✓ The initial ₹1 lakh crore of Central Assistance is strategically designed to act as a multiplier, pulling in market and state funds to unleash a total investment of ₹4 lakh crore in the urban infrastructure sector over the next five years.
 - ✓ Beyond its primary operational window ending in FY 2030–31, the initiative permits an extendable project implementation window stretching up to FY 2033–34.
 - ✓ **Launched to give effect to the urban development visions outlined in Budget 2025–26, the fund focuses on three specific themes:** Cities as Growth Hubs, Creative Redevelopment of Cities, and Water and Sanitation.

- ✓ The entire tracking process is completely paperless, managed through a single digital portal hosted by the Ministry of Housing and Urban Affairs (MoHUA).

✓ Project Verticals:

- **Growth Hubs:** Integrated spatial economic and transit planning, greenfield/semi-greenfield developments, and economic corridors.
- **Creative Redevelopment:** Renewal of central business districts (CBDs), heritage cores, brownfield regeneration, climate resilience, and building countermagnets to decongest existing cities.
- **Water and Sanitation:** Upgrading water grids, sewerage networks, stormwater systems, and integrated solid waste management featuring legacy waste remediation.

- ✓ UCF applies to all Indian cities with a population of 10 lakh or more (2025 estimates), all State and Union Territory capitals, and major industrial cities with a population of 1 lakh or more.
- ✓ For smaller urban local bodies (under 1 lakh population) and cities in Hilly or Northeastern states, the scheme provides a Central guarantee of up to ₹7 crore or 70% of the loan amount for first-time loans (supporting minimum ₹20 crore projects). For subsequent loans, the guarantee becomes ₹7 crore or 50% of the loan amount (supporting minimum ₹28 crore projects).

Q 36. Ans. A

Explanation:

- **Statement A is correct:** The Union Cabinet approved BHAVYA as a fully-funded Central Sector Scheme with a dedicated financial allocation of exactly ₹33,660 crores to build world-class industrial infrastructure.
- **Statement B is correct:** The core objective of the scheme is to systematically develop 100 investment-ready, plug-and-play industrial parks across India to drastically reduce the entry barriers for manufacturing investors.
- **Statement C is correct:** The scheme features a designated implementation window operating over a six-year period, commencing in the financial year 2026-27 and concluding in 2031-32.
- **Statement D is incorrect:** The Bharat Audyogik

Vikas Yojna (BHAVYA) was conceptualised and launched by the Department for Promotion of Industry and Internal Trade (DPIIT), which falls under the Ministry of Commerce and Industry, not the Ministry of Micro, Small and Medium Enterprises.

● **Additional Facts:**

- ✓ The National Industrial Corridor Development Corporation (NICDC) serves as the central Project Management Agency responsible for overseeing and implementing the scheme.
- ✓ The scheme provides central capital assistance of up to ₹1 crore per acre for core, value-added, and social infrastructure, and funds up to 25% of the project cost for external last-mile connectivity to national freight grids.
- ✓ The selected industrial parks will range from 100 to 1,000 acres, with a reduced minimum threshold of 25 acres explicitly permitted for hilly terrains, Northeastern states, and Union Territories.

Q 37. Ans. B

Explanation:

- The “J-curve effect” describes the time path of a country’s trade balance following a depreciation or devaluation of its currency. Because trade contracts are often rigid in the short term, a currency depreciation initially makes imports more expensive and exports cheaper in price, without instantly changing the volume of goods traded. This causes a temporary widening of the trade deficit (the downward hook of the “J”).
- Over time, as consumers and producers adjust to the new relative prices, export volumes rise, and import volumes fall, leading to a long-term improvement in the trade balance (the upward trajectory of the “J”).
- **Recent example:** A similar short-term dynamic was observed with the Indian Rupee, which depreciated by roughly 11% against the USD between March 2025 and March 2026. Initially, India saw no immediate relief in its trade balance. Because India is a structural net importer of crude oil, the import bill remained instantly elevated in Rupee terms, while it took time for foreign buyers to recognise the new relative prices and for domestic export volumes to adjust.

Q 38. Ans. A

Explanation:

- The **Ministry of Commerce and Industry** does not have representation on the Insolvency and Bankruptcy Board of India (IBBI).
- According to Section 189 of the Insolvency and Bankruptcy Code (IBC), 2016, the IBBI board must include exactly three ex officio members representing the Central Government (not below the rank of Joint Secretary). These three members are drawn exclusively from the **Ministry of Finance**, the **Ministry of Corporate Affairs**, and the **Ministry of Law and Justice**.

● **Additional Facts**

- ✓ **Total Composition:** The IBBI board consists of a maximum of 10 members: one Chairperson, three representatives from the specified Central Government ministries, one member nominated by the Reserve Bank of India (RBI), and five other members nominated by the Central Government (of which at least three must be whole-time members).
- ✓ **Establishment:** The IBBI was established on October 1, 2016, as the apex regulatory body implementing the Insolvency and Bankruptcy Code (IBC), 2016.
- ✓ **Unique Dual Role:** It is a unique regulator because it regulates both the profession (Insolvency Professionals, Insolvency Professional Agencies, and Information Utilities) and the processes (Corporate Insolvency Resolution, Corporate Liquidation, and Individual Insolvency/Bankruptcy).
- ✓ **Nodal Ministry:** Although multiple ministries are represented on the board, the IBBI functions administratively under the **Ministry of Corporate Affairs (MCA)**.

Q 39. Ans. C

Explanation:

- **Statement 1 is incorrect:** The platform represents a major targeted fiscal intervention for agritech modernisation, receiving a dedicated total central financial allocation of **₹150 crore** to establish its core AI framework and multi-channel delivery network.
- **Statement 2 is correct:** Structurally, Bharat-VISTAAR is built as a core component of India’s Digital Public Infrastructure (DPI) ecosystem, leveraging data to provide small and marginal farmers with real-time, location-specific, and personalised agricultural solutions.
- **Statement 3 is incorrect:** The platform does

not consolidate state-level cooperative credit architectures. Instead, its current economic and operational focus is the integration of ten major Central Government agricultural schemes, such as PM-KISAN, PMFBY, and the Kisan Credit Card (KCC), to streamline information delivery and service access.

● **Additional Facts:**

- ✓ The acronym VISTAAR stands for Virtually Integrated System to Access Agricultural Resources. It is managed directly under the aegis of the **Ministry of Agriculture & Farmers' Welfare**.
- ✓ To address the digital divide among smallholders, the system provides an Interactive Voice Response System (IVRS) for non-smartphone users via a dedicated short-code number (155261), alongside a voice-based AI chatbot, a mobile application, and a centralised web portal.
- ✓ The advisory engine does not rely on unverified data; it actively synthesises scientific agricultural data and crop management practices validated by the Indian Council of Agricultural Research (ICAR).
- ✓ Reflecting India's evolving data governance architecture, the platform has been developed in strict compliance with the Digital Personal Data Protection (DPDP) Act, 2023, and satisfies all cybersecurity standards set by MeitY and CERT-In.

Q 40. Ans. D

Explanation:

- **The Weighted Average Lending Rate (WALR)** is an aggregate economic indicator that reflects the actual average interest rate a bank is earning across all its outstanding loans, calculated by weighing the interest rates of individual loans against their respective principal amounts.
- **Additional Facts:**
 - ✓ **WADTDR (Weighted Average Domestic Term Deposit Rate):** The liability-side equivalent of WALR. It represents the average interest rate a bank is paying out on all its outstanding domestic term deposits (fixed deposits).
 - ✓ **EBLR (External Benchmark Lending Rate):** Introduced by the RBI in 2019, this mandates banks to link all new floating-rate retail and MSME loans to an external benchmark

(like the RBI's Repo Rate or Treasury Bill yields) to ensure faster and more transparent transmission of policy rate changes.

- ✓ **MCLR (Marginal Cost of Funds-based Lending Rate):** Introduced in 2016, this is the internal benchmark rate below which a bank is not permitted to lend. It is calculated based on the incremental cost of arranging fresh funds for the bank.
- ✓ **Base Rate:** The older internal benchmark regime (used before MCLR). It was calculated using the average cost of funds. Loans taken before 2016 that haven't been transitioned are still pegged to the Base Rate.
- ✓ **Spread:** The difference between a bank's lending rate (like WALR) and its deposit rate (like WADTDR). This represents the bank's core profitability margin, technically known as the Net Interest Margin (NIM).

Q 41. Ans. A

Explanation:

- **Faience** was an artificially produced material in the Harappan civilisation. It was created by grinding quartz or sand into a fine powder, mixing it with a binding agent (like gum), and then shaping and firing the mixture. This process produced a shiny, glassy surface that was typically colored blue or sea green and was used to craft luxury items like beads, bangles, and small vessels.
- **Additional Facts:**
 - ✓ **Natural lapis lazuli:** This is a naturally occurring, highly valued blue stone sourced by the Harappans primarily from Shortughai (Afghanistan).
 - ✓ **Powdered steatite paste:** While steatite (soapstone) paste was widely used by Harappans to make micro-beads and seals, faience is distinctively manufactured through high-temperature firing.

Q 42. Ans. A

Explanation:

- **Statement 1 is correct:** The Satavahanas are historically recognised as the first Indian dynasty to issue epigraphic land grants that included specific administrative and fiscal exemptions. Inscriptions, such as those at Nasik (issued by Gautamiputra Satakarni) and Naneghat, explicitly mention granting lands to Buddhist monks and Brahmanas with immunities, such as exemption from taxes and from the entry of royal troops or officials.

- **Statement 2 is incorrect:** Epigraphic evidence from early rock-cut caves in the western Deccan (like Karle, Bhaja, and Kanheri) reveals that the earliest donations were not primarily from royal patrons. Instead, they were characterised by collective patronage from a wide social base, heavily driven by merchants (gahapatis and sethis), artisans, trade guilds (srenis), and even ordinary monks and nuns.
- **Additional Facts:**
 - ✓ The practice of granting tax-free land initiated by the Satavahanas paved the way for the feudalisation of the Indian economy and administrative system in the post-Mauryan and Gupta periods.
 - ✓ Guilds (srenis) played a highly influential role in the Satavahana economy. They did not just act as trade organisations; they also functioned as rudimentary banks, receiving endowments and paying interest that was often used to perpetually fund religious activities at these cave sites.

Q 43. Ans. C

Explanation:

- **Statement 1 is correct:** The Later Vedic period witnessed a significant shift in religious pantheons. The early Rig Vedic gods like Indra and Agni lost their supreme importance, and Prajapati (the Creator) emerged as the supreme deity, along with Vishnu (the Preserver) and Rudra (the Destroyer).
- **Statement 2 is incorrect:** Varuna was highly revered in the early Rig Vedic period as the upholder of Rita (the cosmic and moral order). However, he did not maintain this position. In the Later Vedic period, his status declined drastically, and he was relegated to being a minor deity associated with water.
- **Statement 3 is correct:** The social status of women declined significantly during the Later Vedic period. While they actively participated in tribal assemblies such as the Sabha and Vidatha during the Rig Vedic phase, they were deprived of this right and barred from attending political assemblies in the Later Vedic phase.
- **Additional Facts:**
 - ✓ Indra (Purandara or breaker of forts) was the most prominent Rig Vedic deity with 250 hymns dedicated to him, followed closely by Agni (the fire god) with 200 hymns.
 - ✓ The Later Vedic period was characterised by the rigidification of the Varna system. What was once a flexible, occupation-based division in the Rig Vedic era transformed into a strict, hereditary caste hierarchy.

Q 44. Ans. B

Explanation:

- **Bead making (a - 3):** Chanhudaro and Lothal were the premier centres for bead production in the Harappan civilisation, characterised by the discovery of specialised workshops and drills.
- **Shell work (b - 1):** Nageshwar and Balakot were coastal settlements specifically dedicated to crafting shell objects, including bangles, ladles, and inlay work, which were then transported to other urban centres.
- **Bronze metallurgy (c - 2):** Harappan smiths mastered the lost-wax (cire perdue) casting technique to create intricate bronze statues, the most famous being the Dancing Girl of Mohenjo-Daro.
- **Faience production (d - 4):** Faience was a highly valued, artificially manufactured synthetic paste made from crushed quartz and a binding agent, used to make luxury items like miniature vessels and ornaments.
- **Additional Facts:**
 - ✓ **Chanhudaro's uniqueness:** Unlike other major Harappan cities like Mohenjo-Daro or Harappa, Chanhudaro was a tiny settlement almost exclusively devoted to craft production, with no evidence of a citadel.
 - ✓ **Materials for beads:** Bead makers utilised a wide variety of materials, including beautiful red carnelian, jasper, crystal, quartz, and steatite (a very soft stone that was easily moulded).

Q 45. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Punch-marked coins, made mostly of silver (and occasionally copper), are the earliest known metallic currency in India. Their introduction during the 6th century BCE revolutionised trade and moved the economy away from the barter system.
- **Pair 2 is correctly matched:** Shrenis were highly organised guilds or corporations of merchants and artisans. They played a vital role in the urban economy by regulating production, fixing wages and prices, and acting as rudimentary banks.
- **Pair 3 is correctly matched:** The period saw the rise of large urban centres known as Mahanagaras. Cities like Rajagriha, Pataliputra, Champa, Sravasti, Varanasi, and Kausambi served as the massive administrative capitals and commercial hubs of the Mahajanapadas.

● **Additional Facts:**

- ✓ **The Iron Revolution:** The Second Urbanisation (6th century BCE onwards) was heavily driven by the widespread use of iron technology. Iron axes helped clear the dense forests of the Gangetic plains, and the iron ploughshare led to a massive agricultural surplus, which was essential to sustain large urban populations.
- ✓ **First vs. Second Urbanisation:** While the First Urbanisation (Harappan civilisation) was centred in the Indus Valley and was characterised by bronze technology, the Second Urbanisation was centred in the middle Gangetic plains and was characterised by iron technology.
- ✓ **Ceramic Marker:** The defining pottery of this era is the Northern Black Polished Ware (NBPW), a glossy, high-quality ceramic often associated with the wealthy urban elite.

Q 46. Ans. B

Explanation:

- Northern Black Polished Ware (NBPW) is universally recognised as the diagnostic ceramic marker for the Second Urbanisation in the Gangetic plains (c. 6th century BCE onwards). It is a highly lustrous, glossy, and fine-quality luxury pottery that reflects the technological advancements and growing urban wealth of the Mahajanapada and Mauryan periods.
- **Additional Facts:**
 - ✓ **Painted Grey Ware (PGW):** This pottery is closely associated with the Later Vedic period and the early phase of iron use in northern India, preceding the Second Urbanisation phase.
 - ✓ **Ochre Coloured Pottery (OCP):** This is a much earlier ceramic tradition primarily associated with the Chalcolithic culture of the Ganga-Yamuna Doab region, often found in association with Copper Hoards.
 - ✓ **Black and Red Ware (BRW):** While found across various periods—including the Chalcolithic phase and the Megalithic culture of South India—it is not the specific, defining marker of the Second Urbanisation.

Q 47. Ans. A

Explanation:

- **Both the assertion and the reason are true, and the reason correctly explains the assertion:** During the early Rig Vedic period, the political structure was predominantly tribal (Jana), and the king was viewed mainly as a protector or leader of the

tribe. However, during the Later Vedic period, the transition to settled agriculture led to the formation of distinct territorial states (Janapadas). As the kings expanded their domains and sought absolute control over these territories, they needed to justify their elevated status and distance themselves from ordinary tribespeople. To achieve this divine legitimacy, they formed a strong alliance with the Brahmanas and sponsored highly elaborate, public rituals that visually and spiritually validated their supreme royal authority.

● **Additional Facts:**

- ✓ **Rajasuya (Royal Consecration):** A complex ritual that supposedly conferred supreme power and divine attributes upon the king.
- ✓ **Ashvamedha (Horse Sacrifice):** A ritual meant for territorial expansion. A royal horse was let loose to wander; if it passed through another chief's territory unchallenged, that land was claimed by the king.
- ✓ **Vajapeya (Chariot Race):** A ritual where the king's chariot was deliberately made to win a race against his kinsmen, symbolically establishing his unquestioned superiority and rejuvenating his power.

Q 48. Ans. C

Explanation:

- The Gandhara School of Art flourished in the northwestern frontier of the Indian subcontinent, heavily patronised by the **Kushana** emperors (especially Kanishka). Its strategic location along the Silk Road facilitated intense cultural interaction between Indian, Greco-Roman, and Central Asian traditions. This resulted in a unique syncretic art style that combined Buddhist themes with Hellenistic (Greek) and Roman aesthetic features.
- **Additional Facts:**
 - ✓ **Foreign Influence vs. Isolation:** The region was far from isolated; it was a major cultural melting pot. The Gandhara Buddha statues often feature wavy hair, athletic bodies, and heavy, toga-like garments, directly inspired by Greco-Roman gods like Apollo.
 - ✓ **Gupta Patronage:** The Gupta rulers primarily patronised the Mathura and Sarnath schools of art, not the Gandhara school. Mathura art, unlike Gandhara, was completely indigenous.
 - ✓ **Religion:** The Gandhara School was almost exclusively dedicated to Mahayana Buddhism, not Jainism.

- ✓ **Material:** The primary material used in early Gandhara art was bluish-grey sandstone (schist), while later phases increasingly utilised stucco and mud.

Q 49. Ans. C

Explanation:

- **Statement 1 is correct:** In an ambitious economic experiment, Muhammad bin Tughlaq introduced a token currency system. He issued copper and brass coins and decreed that their face value would be strictly identical to that of the standard silver tanka.
- **Statement 3 is correct:** To recover from agricultural distress and famines, he established a specialised department of agriculture known as the Diwan-i-Amir-Kohi. Its primary aim was to bring uncultivated land under the plough, provide state loans (sondhar) to farmers, and encourage the transition from inferior to superior crops.
- **Statement 2 is incorrect:** The capital was shifted back from Daulatabad to Delhi not because of a sudden Mongol invasion, but due to a strategic realisation. The Sultan realised that just as it was difficult to control the southern provinces from Delhi, it was equally impossible to control the northern frontiers (which were always vulnerable) and manage rebellions in the north from Daulatabad.
- **Additional Facts:**
 - ✓ **Failure of the Token Currency:** The token currency experiment failed miserably because the state did not monopolise the minting of coins. Every goldsmith's house became a mint, leading to widespread forgery. The Sultan eventually had to withdraw the project and exchange genuine silver coins for the forged copper ones, severely draining the treasury.
 - ✓ **Karachil and Khurasan Expeditions:** Muhammad bin Tughlaq is also known for his highly ambitious but flawed military campaigns. He raised and paid a massive army in advance for an invasion of Khurasan (Central Asia), which was ultimately abandoned. His Karachil expedition (in the Kumaon hills) suffered heavy casualties due to the harsh winter and difficult terrain.

Q 50. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Alauddin Khalji introduced the Dagh (branding) system for cavalry horses. This ensured that high-quality horses were

not replaced by inferior ones after being passed in the muster, thereby preventing corruption and maintaining the cavalry's strength.

- **Pair 2 is correctly matched:** The Chehra (or Huliya) system was a detailed descriptive roll maintained for every individual soldier. This prevented the substitution of actual soldiers with proxies or untrained men during military inspections.
- **Pair 3 is incorrectly matched:** The Diwan-i-Mustakharaj was not associated with market control; it was a special revenue department created by Alauddin to enquire into the revenue arrears and collect them strictly from the revenue collectors (Amils and Muqaddams).
- **Additional Facts:**
 - ✓ **Market Control:** The price control and market regulations in Delhi were actually managed by a separate department headed by the Diwan-i-Riyasat, under whom officers known as Shahna-i-Mandi (market superintendents) operated to ensure strict compliance with fixed prices.
 - ✓ **Standing Army:** Alauddin Khalji was the first Sultan of Delhi to lay the foundation of a permanent, centralised standing army, paid fully in cash (Naqad), rather than relying on the troops supplied by nobles or assigning Iqtas (land grants) in lieu of salary.

Q 51. Ans. A

Explanation:

- Under Article 72, the President's pardoning power extends to sentences by a court martial, sentences of death, and punishment for an offence against a Union law, and it is wider than the Governor's power under Article 161, which does not extend to a death sentence or to a court martial sentence.
- The forms of clemency available under Article 72 include pardon, which removes both the sentence and the conviction; commutation, which substitutes a lighter form of punishment; remission, which reduces the period without changing its character; respite, which awards a lesser sentence on special grounds; and reprieve, which is a temporary stay of execution.

Q 52. Ans. B

Explanation:

- **Assertion is true:** Under Article 109, a money bill can be introduced only in the Lok Sabha and not in the Rajya Sabha, reflecting the principle that the popular House has primacy over financial legislation.

- Reason is true but does not correctly explain the **Assertion**: It is indeed the Speaker of the Lok Sabha who certifies whether a bill is a money bill, and this certification is final, but this certifying function explains how a bill's character as a money bill is conclusively determined, not why a money bill can be introduced only in the Lok Sabha; the restriction on the place of introduction flows directly from Article 110 itself, independent of the certification power.
- Once passed by the Lok Sabha, a money bill is transmitted to the Rajya Sabha, which can only make recommendations within fourteen days and cannot amend or reject it; if the Rajya Sabha does not return the bill within that period, it is deemed to have been passed by both Houses in the form originally passed by the Lok Sabha.

Q 53. Ans. C

Explanation:

- **Speaker of the Lok Sabha** is not appointed by the President of India. According to **Article 93** of the Indian Constitution, the Speaker is **elected** by the members of the Lok Sabha from amongst themselves as soon as possible after the House is constituted.
- **Attorney General of India**: Appointed by the President of India under **Article 76(1)**.
- **Comptroller and Auditor General of India (CAG)**: Appointed by the President of India under **Article 148(1)** by warrant under his hand and seal.
- **Chief Election Commissioner**: Appointed by the President of India under **Article 324(2)**, subject to the provisions of any law made by Parliament.

Q 54. Ans. B

Explanation:

- The 44th Amendment Act of 1978 deleted the right to property from the list of fundamental rights by repealing Article 19(1)(f) and Article 31, and inserted a new Article 300A, providing that no person shall be deprived of property save by authority of law, thereby converting it into an ordinary legal right.
- This change was made mainly to remove the obstacles the fundamental right to property had created for land reform and other socioeconomic legislation. The 42nd Amendment Act of 1976 had earlier added the Fundamental Duties and several new directive principles, the 24th Amendment Act of 1971 had affirmed Parliament's power to amend fundamental rights, and the 86th Amendment Act of 2002 inserted Article 21A on the right to education.

Q 55. Ans. D

Explanation:

- **Statement 1 is correct (ADM Jabalpur v. Shivkant Shukla, 1976)**: Often referred to as the Habeas Corpus case, the Supreme Court ruled during the 1975 Emergency that the right of any person to move any court for the enforcement of the rights conferred by Article 21 could be suspended. The widespread criticism of this judgment directly catalysed the **44th Constitutional Amendment Act (1978)**, which amended Article 359 to explicitly state that the enforcement of fundamental rights under Articles 20 (protection in respect of conviction for offences) and 21 (protection of life and personal liberty) cannot be suspended even during a National Emergency.
- **Statement 2 is correct (Kesavananda Bharati v. State of Kerala, 1973)**: In a historic 7-6 majority decision, a 13-judge constitutional bench ruled that while Parliament has wide powers to amend the Constitution under Article 368, it cannot alter, destroy, or abrogate its "Basic Structure" or fundamental framework. This established the **Basic Structure Doctrine**.
- **Statement 3 is correct (S.P. Gupta v. Union of India, 1981)**: Also known as the First Judges Case, this judgment was instrumental in formalising **Public Interest Litigation (PIL)** in India. Justice P.N. Bhagwati articulated the relaxation of the traditional, rigid rule of locus standi (the right to bring an action), allowing any public-spirited citizen or social action group to approach the court on behalf of those who, due to poverty, ignorance, or social/economic disability, cannot approach the courts themselves.
- **Statement 4 is correct (I.R. Coelho v. State of Tamil Nadu, 2007)**: A 9-judge bench unanimously held that laws placed in the **Ninth Schedule** (which traditionally enjoyed blanket immunity from judicial review) on or after **April 24, 1973** (the date of the Kesavananda Bharati judgment) are subject to judicial review. If such laws violate the fundamental rights forming part of the Basic Structure (specifically testing against the "Golden Triangle" of Articles 14, 19, and 21), they can be struck down.

Q 56. Ans. C

Explanation:

- **Pair 1 is correctly matched (Judge of the Supreme Court)**: Pursuant to **Article 124(2)** of the Constitution of India, a judge of the Supreme

Court holds office until attaining the age of **sixty-five years**.

- **Pair 2 is incorrectly matched (Judge of a High Court):** Under **Article 217(1)** of the Constitution, a judge of a High Court holds office until attaining the age of **sixty-two years**. Although legislative proposals and law commission reports have historically recommended aligning this with the Supreme Court's retirement age, the constitutional limit remains sixty-two.
- **Pair 3 is correctly matched (Comptroller and Auditor General):** As prescribed under Section 4 of the Comptroller and Auditor-General's (Duties, Powers and Conditions of Service) Act, 1971 (enacted under the enabling provisions of **Article 148(3)**), the CAG holds office for a term of **six years** or until attaining the age of **sixty-five years**, whichever occurs earlier.
- **Pair 4 is incorrectly matched (Election Commissioner):** According to Section 9 of The Chief Election Commissioner and other Election Commissioners (Appointment, Conditions of Service and Term of Office) Act, 2023 (governed under **Article 324**), both the Chief Election Commissioner (CEC) and other Election Commissioners (ECs) hold office for a tenure of **six years** or until they attain the age of **sixty-five years**, whichever is earlier.

Q 57. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The "None of the Above" (NOTA) option was officially introduced on Electronic Voting Machines (EVMs) following the landmark Supreme Court judgment in *People's Union for Civil Liberties (PUCL) v. Union of India* (2013). The Court ruled that the right to register a negative vote is a fundamental right of a voter under **Article 19(1)(a)** (freedom of speech and expression), directing the Election Commission of India (ECI) to facilitate this provision.
- **Pair 2 is incorrectly matched:** The Model Code of Conduct (MCC) **does not** have direct statutory backing under any specific Act of Parliament. It is a consensus-based framework voluntarily agreed upon by political parties to maintain campaign decorum. The ECI enforces the MCC using its inherent constitutional power of superintendence, direction, and control of elections under **Article 324**. While certain violations of the MCC (such as bribery or hate speech) can be prosecuted under corresponding sections of the *Bharatiya Nyaya*

Sanhita (formerly IPC) or the Representation of the People Act, 1951, the MCC document itself is not a statutory enactment.

- **Pair 3 is correctly matched:** Under **Article 329(a)** of the Constitution, read with Section 10(2) of the Delimitation Act, 2002, the orders of the Delimitation Commission have the force of law and **cannot be questioned in any court**. This constitutional bar is designed to prevent electoral delays that would arise from endless litigation over constituency boundaries. Furthermore, when these orders are laid before the Lok Sabha and the respective State Legislative Assemblies, the legislatures cannot effect any modifications to them.
- **Pair 4 is correctly matched:** A by-election (or bye-election) is statutorily defined as an election held to fill a **casual vacancy**. According to Sections 147 through 151 of the Representation of the People Act, 1951, a casual vacancy arises when an elected member's seat becomes vacant before the expiration of their term due to death, resignation, or disqualification. The ECI is mandated to fill such vacancies through a by-election within six months of the vacancy occurring.

Q 58. Ans. B

Explanation:

- The GST Council, established under **Article 279A**, has the Union Finance Minister as its ex officio Chairperson, with the Union Minister of State for Finance or Revenue and the state ministers in charge of finance or taxation as its other members, and decisions require a majority of not less than three-fourths of the weighted votes cast.
- This is distinct from the Finance Commission, whose chairman is a separately selected person with experience in public affairs, from NITI Aayog, which is chaired by the Prime Minister, and from the Public Accounts Committee, a parliamentary committee traditionally chaired by a member of the opposition and not by the Finance Minister.

Q 59. Ans. C

Explanation:

- **Pair 1 is incorrectly matched:** The Goods and Services Tax (GST) Council is a constitutional body, but it is established under **Article 279A**, which was inserted by the 101st Constitutional Amendment Act, 2016. **Article 258A**, conversely, pertains to the power of the States to entrust functions to the Union.
- **Pair 2 is correctly matched:** The Finance

Commission is a constitutional body established under **Article 280** of the Constitution. It is constituted periodically by the President of India to recommend the distribution of tax revenues between the Union and the States and amongst the States themselves.

- **Pair 3 is incorrectly matched:** The National Institution for Transforming India (NITI Aayog) is **not a constitutional body**. It was established on January 1, 2015, via an executive resolution of the Union Cabinet, replacing the erstwhile Planning Commission. Article 338 actually provides for the establishment of the National Commission for Scheduled Castes (NCSC).
- **Pair 4 is correctly matched:** The NCBC was elevated to constitutional status via the 102nd Constitutional Amendment Act, 2018, which inserted **Article 338B** into the Constitution. Prior to this amendment, it functioned as a statutory body under the Ministry of Social Justice and Empowerment.

Q 60. Ans. B

Explanation:

- **Statement 1 is incorrect:** The First Administrative Reforms Commission was established in January 1966 and was initially chaired by Morarji Desai. It was only after Desai became the Deputy Prime Minister of India in 1967 that he resigned from the commission, and K. Hanumanthaiya subsequently took over as its chairman.
- **Statement 2 is correct:** The ARC is an ad-hoc committee appointed by the Government of India via an executive resolution to review the public administration system. Consequently, its reports and policy proposals are advisory in nature. The government exercises executive discretion in accepting, modifying, or rejecting its recommendations.
- **Statement 3 is incorrect:** The Second Administrative Reforms Commission (constituted in 2005 under the chairmanship of Veerappa Moily) is not a constitutional body, nor is it an adjudicatory tribunal. Article 323A pertains to the establishment of Administrative Tribunals (like the Central Administrative Tribunal) by Parliament for the adjudication of disputes relating to public services. The ARC, conversely, is an executive policy-review committee.
- **Statement 4 is correct:** A foundational conceptual contribution of the First ARC was its comprehensive study on the machinery for the redressal of citizens' grievances. In its 1966 interim report, it explicitly recommended the establishment of two special

authorities, the Lokpal at the central level and the Lokayukta at the state level, modelled after the Scandinavian institution of the Ombudsman.

Q 61. Ans. C

Explanation:

- **Asia's largest grain-based ethanol plant** is being established at **Jamui, Bihar**, by **Ankur Biochem**. The plant will have a production capacity of **750 kilolitres per day (KLPD)** and aims to strengthen **green energy production** by converting surplus agricultural produce such as **maize and rice** into **ethanol (C₂H₅OH)**. The project is expected to promote biofuel production, enhance value addition to agricultural produce, and support farmers while contributing to India's clean energy and ethanol blending initiatives.

Q 62. Ans. B

Explanation:

- Recent studies have found **high concentrations of phthalates** in the **soil of Patna**. Phthalates are **plasticisers** commonly used to make plastics **soft and flexible**. Improperly discarded plastic waste gradually degrades, especially when temperatures exceed **40°C**, releasing these chemicals into the soil through a process known as **leaching**. The most commonly detected and harmful phthalate is **DEHP (Di(2-ethylhexyl) phthalate)**. Once released, these chemicals can enter the **food chain** and may disrupt the **hormonal (endocrine) system**, posing health risks to both humans and animals. During rainfall, phthalates can also be carried into **groundwater** and eventually the **Ganga River**, increasing the risk of wider environmental contamination.

Q 63. Ans. B

Explanation:

- The **7th Bihar State Finance Commission** is chaired by **Ashok Kumar Chaudhary**, a **retired IAS officer**. The Commission is responsible for reviewing the financial position of **Panchayati Raj Institutions (PRIs)** and **Urban Local Bodies (ULBs)** and recommending the principles for the **equitable distribution of financial resources** between the **State Government** and local bodies, thereby promoting **fiscal decentralization**. The **State Finance Commission (SFC)** is a constitutional body constituted under **Articles 243-I and 243-Y** of the **Constitution of India**.

Q 64. Ans. D**Explanation:**

- The first phase of the Patna Metro Rail Service was inaugurated by Bihar Chief Minister Nitish Kumar on **6 October 2025**. On the occasion, he travelled on the inaugural route from the **New Pataliputra Bus Terminal (ISBT)** to **Bhootnath Station**. The operational stretch, known as the **Priority Corridor**, is an **approximately 3.6-km elevated metro corridor** connecting ISBT and Bhootnath Station. The inauguration marked the commencement of metro services in Bihar after nearly **six years** of construction. The **Patna Metro Rail Project** was officially launched by **Prime Minister Narendra Modi** in **February 2019**.

Q 65. Ans. C**Explanation:**

- The **Flood Management Improvement Support Centre (FMISC)** under the **Water Resources Department, Government of Bihar**, has identified **eight major river basins** as priority areas for flood monitoring and management. These are the **Gandak, Burhi Gandak, Bagmati, Adhwara Group, Kamla Balan, Kosi, Mahananda, and the Ganga Stem (within Bihar)**. Together, these river basins represent the **most flood-prone regions** of the state. The **Flood Management Information System (FMIS)** uses this hydrographic framework to monitor real-time rainfall, generate daily flood alerts, and run advanced digital forecasting models to predict flood movement from upstream catchment areas, thereby strengthening flood preparedness and disaster management.

Q 66. Ans. A**Explanation:**

- The **Kosi River** is described by the **Government of Bihar** as a transboundary river that drains the **northern slopes of Tibet** and the **southern slopes of Nepal** before entering India. Flowing through **China (Tibet), Nepal, and Bihar**, it is one of the most dynamic rivers of the Himalayan region. In Nepal, the river is known as the **Saptakoshi** because it is formed by the confluence of **seven major tributaries**, including the **Arun and Sun Kosi**, both of which originate in Tibet. Owing to its **heavy sediment load** and frequent channel shifts, the Kosi is popularly known as the **“Sorrow of Bihar”**, as it causes recurrent floods across the state’s northern

plains. After entering Bihar, the river spreads into several distributaries before finally joining the **Ganga River near Kursela in Katihar district**.

Q 67. Ans. A**Explanation:**

- **Statement 1 is correct:** Bihar is divided into four agro-climatic zones for agricultural planning. **Agro-Climatic Zone IIIA (South-East Alluvial Plain)** comprises six districts—**Bhagalpur, Banka, Munger, Jamui, Lakhisarai, and Sheikhpura**. These districts have similar soil, rainfall, and climatic characteristics, which form the basis for region-specific agricultural planning and crop recommendations. Therefore, Statement 1 is correct.
- **Statement 2 is incorrect:** Patna does not belong to **Agro-Climatic Zone IIIA**. Instead, it is part of **Agro-Climatic Zone IIIB (South-West Alluvial Plain)** along with districts such as Bhojpur, Buxar, Rohtas, Kaimur, Nalanda, Gaya, Aurangabad, Nawada, Jehanabad, and Arwal. Hence, placing Patna in Zone IIIA is factually incorrect.

Q 68. Ans. A**Explanation:**

- **Statement 1 is correct:** **Bihar Rajya Beej Nigam Limited (BRBN)** was established by the Government of Bihar to ensure the availability of high-quality seeds to farmers across the state. Its core functions include the production, processing, certification support, storage, marketing, and distribution of foundation and certified seeds under various government schemes. The corporation also facilitates doorstep delivery of seeds and supports the implementation of the National Seed Project in Bihar.
- **Statement 2 is correct:** BRBN is a **Government of Bihar undertaking** that functions **under the Department of Agriculture**. It operates from Krishi Bhawan, Patna, and implements seed-related programmes and schemes of the Agriculture Department to improve seed replacement rates and agricultural productivity in the state.
- **Statement 3 is incorrect:** BRBN has **no role in regulating fertiliser prices**. Its mandate is confined to seed production, processing, marketing, and distribution. Fertiliser pricing and regulation are governed separately through fertiliser-related policies and the concerned government authorities, not by Bihar Rajya Beej Nigam Limited.

Q 69. Ans. B**Explanation:**

- **Assertion (A) is true:** Bihar agriculture follows three major cropping seasons—Kharif, Rabi, and Zaid. This classification is based on the **cropping calendar, sowing time, harvesting period, rainfall availability, and temperature requirements of crops.**
- **Reason (R) is true:** Bihar has suitable climatic conditions, including monsoon rainfall, fertile alluvial soil, irrigation facilities, and suitable temperature conditions, which allow agricultural activities in different seasons.
- The Reason does **not explain** the Assertion. The existence of three agricultural seasons is not because cultivation is possible throughout the year. Even if cultivation is possible year-round, the seasons are divided according to **crop cycles and climatic requirements of individual crops.**

Q 70. Ans. A**Explanation:**

- During the **Home Rule Movement**, Annie Besant paid a brief but significant visit to **Patna on 18 April 1918**. She returned to Bihar on **25 May 1918** for another short visit to strengthen the Home Rule campaign, mobilise public support, and coordinate with local political leaders. During her stay in Patna, she was welcomed and hosted by prominent leaders of the **Bihar branch of the Home Rule League**, including **Maulana Mazharul Haque**, whose support played an important role in expanding the movement's influence in the province.

Q 71. Ans. C**Explanation:**

- The **27th Session of the Indian National Congress** was held at **Bankipore (Patna), Bihar**, from **26–28 December 1912**, marking the **first Congress session to be hosted in Bihar**. The session was **presided over by Raghunath Narasinha Mudholkar** and is remembered as the **first Indian National Congress convention attended by Jawaharlal Nehru**. **Maulana Mazharul Haque** served as the **Chairman of the Reception Committee**, while **Sachchidanand Sinha** was the **General Secretary**. Both leaders played significant roles in Bihar's political development—**Mazharul Haque** later emerged as a key figure in the **Champaran Satyagraha** and founded the **Sadaqat Ashram**, whereas **Sachchidanand Sinha** made notable contributions to the creation of the separate province of Bihar.

Q 72. Ans. C**Explanation:**

- **Legendary folk singer Sharda Sinha** was **posthumously awarded the Padma Vibhushan, India's second-highest civilian honour, in 2025**. Popularly known as "**Bihar Kokila**," she was honoured for her outstanding contribution to the preservation and promotion of **Bhojpuri and Maithili folk music**. The **Padma Awards**—comprising the **Padma Vibhushan, Padma Bhushan, and Padma Shri**—do not carry any cash prize, financial allowance, or monetary grant. Awardees are instead presented with a **Padma medallion** featuring a lotus motif and a **Sanad (certificate)** signed by the **President of India**.

Q 73. Ans. B**Explanation:**

- **Statement 1 is incorrect:** **Madhushravani** is **not associated with the Anga region**. It is a traditional festival of the **Mithilanchal (Mithila) region of Bihar**. The festival is celebrated mainly in areas associated with Mithila culture, such as **Madhubani, Darbhanga** and surrounding regions.
- **Statement 2 is correct:** **Madhushravani** is mainly observed by **newly married women/brides**. During this festival, newlywed women perform rituals and worship associated with **Naga devatas (serpent deities), Gauri, and other traditional deities** for marital happiness and well-being. The Bihar Tourism Department also mentions that the festival is observed by married women and has special traditions related to newly married girls.

Q 74. Ans. B**Explanation:**

- **Statement 1 is incorrect:** **Patna Kalam** (also known as **Company Painting of Patna**) did **not originate from the Pahari School of Painting**. It developed as a distinct style in **Patna, Bihar**, during the late 18th and 19th centuries. It emerged mainly from the **Mughal miniature painting tradition** after many Mughal artists migrated to eastern India following the decline of Mughal patronage. Therefore, linking its origin with the Pahari School is incorrect.
- **Statement 2 is correct:** A unique feature of **Patna Kalam** was its focus on the **everyday life of ordinary people**. Unlike many traditional Indian painting schools that focused on royal courts, mythology, or religious themes, **Patna Kalam** artists painted

subjects such as artisans, craftsmen, festivals, occupations, markets, local customs, and social life.

- **Statement 3 is correct:** Patna Kalam flourished under the patronage of **British East India Company officials**, including civil and military officers stationed in Bihar. These officials commissioned paintings depicting Indian society, occupations, and local scenes, which contributed to the growth of this style.

Q 75. Ans. D

Explanation:

- “**The Partition of the Lower Provinces – An Alternative Proposal**” was an influential article published in **1906**, jointly authored by **Mahesh Narayan** and **Sachchidananda Sinha**. In the article, the two leaders criticised the British plan for the **Partition of Bengal** and proposed an alternative administrative arrangement. They argued that **Bihar**, with its distinct **linguistic, cultural, and historical identity**, should be separated from Bengal and constituted as an independent province. Instead of dividing the Bengali-speaking population, they advocated the creation of a **separate province for the people of Bihar**, a demand that later contributed to the formation of the **Province of Bihar and Orissa** in **1912**.

Q 76. Ans. A

Explanation:

- The **Hemant Trophy** is Bihar’s premier **senior men’s domestic cricket tournament**, organized by the **Bihar Cricket Association (BCA)**. District teams are selected through local training camps across the state. The competition is played in the **50-over (One-Day)** format using a **white ball** and is conducted in **Hemant ‘A’** and **Hemant ‘B’** divisions.

Q 77. Ans. C

Explanation:

- **Raja Man Singh** was appointed the **Subahdar (Governor)** of Bihar by **Emperor Akbar** in **1587 AD**. During his tenure, he consolidated Mughal authority by suppressing **Afghan rebels** and rebellious **zamindars**, compelling influential chiefs such as **Puran Mal of Gidhaur** to acknowledge Mughal rule. Under his administration, Bihar emerged as a secure strategic base for the Mughal Empire’s eastward expansion. In **1590**, Man Singh launched campaigns into **Bengal and Odisha** from Bihar to confront **Qutlu Khan Lohani**, ultimately

securing Mughal supremacy through a combination of military operations and diplomatic agreements with local rulers.

Q 78. Ans. C

Explanation:

- **Statement 1 is correct:** **Padri Ki Haveli**, officially known as **Church of the Virgin Mary**, was established by **Roman Catholic missionaries** in Patna. The present structure was built in the 18th century by the **Capuchin missionaries**, who were part of the Roman Catholic missionary tradition. It is one of the important Christian heritage sites of Bihar.
- **Statement 2 is incorrect:** Padri Ki Haveli is indeed **one of the oldest churches in Bihar**, but it is **located in Patna district**, specifically in the **Patna City area (Padri Ki Haveli locality)**, not in Bhagalpur district. Therefore, the statement is partially correct but becomes incorrect.
- **Statement 3 is correct:** Before beginning her humanitarian work in Kolkata, **Mother Teresa** came to **Patna** and received her basic medical/nursing training at **Holy Family Hospital, Patna**, which is associated with the same Christian missionary tradition in the city. Her connection with Patna is an important part of Bihar’s Christian heritage.

Q 79. Ans. D

Explanation:

- The **Mukhyamantri Kalakar Pension Yojana** is a welfare initiative of the **Government of Bihar** aimed at providing financial security to senior artists. Under the scheme, **eligible veteran artists receive a monthly pension of ₹3,000** to support them in their old age. To avail the benefit, an applicant must be a **permanent resident of Bihar**, **50 years of age or above**, and have **at least 10 years of experience in classical, visual, or performing arts**. Additionally, the applicant’s **annual family income must not exceed ₹1.20 lakh**, and they must not be receiving **any other government pension**.

Q 80. Ans. D

Explanation:

- All the places mentioned are important tourist and religious sites located in the **Munger district** of Bihar.
- **Sita Kund (Kalyan Chak)** – Situated about **6 km east of Munger town**, it is a famous hot spring associated with **Goddess Sita**. The site also includes

Ram Kund, Lakshman Kund, Bharat Kund, and Shatrughan Kund, making it an important Hindu pilgrimage centre.

- **Gurdwara Pakki Sangat (Piparpanti)** – This historic Sikh shrine commemorates the stay of **Guru Tegh Bahadur**, the ninth Sikh Guru, during his journey to Assam. It was here that he received the news of the birth of **Guru Gobind Singh**.
- **Pir Shah Nafah Shrine** – Located inside the **Munger Fort**, it is regarded as the oldest structure within the fort complex. According to local tradition, the saint came from **Persia** via **Ajmer** under the guidance of **Khwaja Moinuddin Chishti** and spent many years in Munger.
- **Sri Krishna Vatika** – Named after **Dr. Sri Krishna Sinha**, the first Chief Minister of Bihar, this landscaped garden lies opposite **Kashtaharini Ghat** and offers scenic views of the **Ganga River**.
- **Chandika Sthan** – A renowned temple dedicated to **Goddess Chandi**, it is traditionally regarded as one of the **64 Shakti Peethas**, where, according to local belief, the **eyes of Goddess Sati** fell.
- **Kashtaharini Ghat** – Located near **Munger Fort**, this sacred ghat is associated with **Lord Rama and Lakshmana**, who are believed to have rested here. Devotees take a holy dip believing it removes suffering and sins.

Q 81. Ans. B

Explanation:

- The 10th edition of Swachh Survekshan (2025–26), launched by the Ministry of Housing and Urban Affairs (MoHUA), has officially designated the theme as “**Swacchata Ki Nayi Pehel- Badhayein Haath, Karein Safai Saath**” (A new initiative for cleanliness—extend your hands, let’s clean together). This theme strongly emphasises community-led cleanliness initiatives, collective responsibility, and round-the-year active citizen participation to sustain the goals of the Swachh Bharat Mission.
- **Additional Facts:**
 - ✓ **Decade Milestone:** The 2025–26 edition marks a decade (10th anniversary) of India’s Swachh Survekshan, which began in 2016 with just 73 cities and has grown into the world’s largest urban sanitation survey assessing thousands of Urban Local Bodies (ULBs).
 - ✓ The 10th edition has expanded its scope from assessing only Ganga towns to including all river towns across the country. It has also introduced a separate matrix for coastal areas

to better address their unique environmental challenges.

- ✓ **Mentorship Programme:** MoHUA introduced a new award category to recognize the Swachh Shehar Jodi (SSJ), a mentorship framework where high-performing mentor cities guide mentee cities in urban waste management and best practices.

Q 82. Ans. C

Explanation:

- **Statement 1 is correct:** The Indian Army has officially declared 2026 as the ‘Year of Networking and Data-Centricity’. This initiative emphasizes the digital transformation of the armed forces with a strong focus on AI-driven warfare, cyber resilience, and integrated battle management systems to enhance real-time decision-making and combat effectiveness.
- **Statement 2 is correct:** ICGS Samudra Pratap, India’s first indigenously designed and built pollution control vessel (PCV), was commissioned into the Indian Coast Guard by Defence Minister Rajnath Singh in January 2026 at Goa Shipyard Limited. Built with over 60% indigenous content, the vessel is equipped with advanced systems to detect, contain, and neutralize marine oil spills and pollution.

Q 83. Ans. C

Explanation:

- **Statement 1 is correct:** The Pradhan Mantri Jan Dhan Yojana (PMJDY) achieved a major milestone in 2025 by successfully crossing 55 crore beneficiary accounts (reaching approximately 55.14 crore by March 2025), reflecting its massive scale in driving financial inclusion across India.
- **Statement 2 is incorrect:** PMJDY has a deeply rooted rural focus, with approximately 67 per cent (around 36.6 crore) of the total registered accounts located in rural and semi-urban areas.
- **Statement 3 is correct:** According to recent data highlighted in the Economic Survey 2025-26, the total disbursements under the Pradhan Mantri Mudra Yojana (PMMY) to micro and small enterprises surpassed Rs. 36.18 lakh crore across more than 55 crore loan accounts, significantly boosting grassroots entrepreneurship.
- **Additional Facts:**
 - ✓ **Women’s Financial Inclusion:** A recognised feature of PMJDY is its focus on gender inclusivity, with women constituting over

55 per cent (more than 30 crore) of the total account holders.

- ✓ **RuPay Integration:** To promote digital banking and a cashless economy, over 37 crore free RuPay debit cards have been issued to Jan Dhan account holders, which also come with an inbuilt accident insurance cover.
- ✓ **PMMY Categories:** Mudra loans are categorised into three segments to signify the growth and funding needs of the beneficiary's micro-business: Shishu (up to Rs. 50,000), Kishore (Rs. 50,000 to Rs. 5 lakh), and Tarun (Rs. 5 lakh to Rs. 10 lakh).

Q 84. Ans. D

Explanation:

- **SHINE App (a - 3):** Indian Railways introduced the Sexual Harassment Incident Notification for Empowerment (SHINE) app to empower women employees and provide a digital platform to report workplace harassment confidentially.
- **Pankhuri Portal (b - 1):** Launched by the Ministry of Women and Child Development, this portal aims to enhance facilities at Anganwadi centres and provide immediate, digitalised solutions to ground-level problems.
- **E-Swasthya Dham Portal (c - 2):** The Government of Uttarakhand launched this portal to facilitate real-time health monitoring and emergency medical tracking for pilgrims participating in the high-altitude Char Dham Yatra.
- **Nivesh Mitra 3.0 (d - 4):** The Government of Uttar Pradesh launched this upgraded, AI-powered single-window clearance portal to simplify industrial approval processes, minimise compliance burdens, and attract corporate investments.

Q 85. Ans. B

Explanation:

- **Global Hunger Index (a - 4):** The Global Hunger Index is a peer-reviewed annual report, jointly published by Concern Worldwide and the German organisation Welthungerhilfe. It is designed to comprehensively measure and track hunger at the global, regional, and national levels.
- **Index of Economic Freedom (b - 2):** Published annually by The Heritage Foundation, this index measures the degree of economic freedom in various nations based on parameters like the rule of law, government size, regulatory efficiency, and open markets.

- **Asia Power Index (c - 3):** Produced by the Lowy Institute, this index measures resources and influence to rank the relative power of states in the Indo-Pacific region.
- **Climate Change Performance Index (d - 1):** The CCPI is an independent monitoring tool published annually by Germanwatch, the NewClimate Institute, and the Climate Action Network (CAN) to track countries' climate protection performance.

Q 86. Ans. B

Explanation:

- The main objective of the PM-PRANAM (PM Programme for Restoration, Awareness, Nourishment and Amelioration of Mother Earth) scheme is to incentivise States and Union Territories to promote the balanced use of chemical fertilizers and encourage the adoption of alternative fertilizers (like organic and bio-fertilizers). By doing so, the scheme aims to reduce the overall consumption of chemical fertilizers, which in turn helps improve soil health, mitigate environmental degradation, and significantly reduce the government's massive fertilizer subsidy burden.
- Under the scheme, 50% of the fertilizer subsidy saved by a State/UT, by reducing chemical fertiliser consumption compared to their previous three-year average, is passed back to that State/UT as a grant to fund further sustainable agriculture initiatives.

Additional Facts:

✓ Initiatives that complement PM-PRANAM:

- **GOBARDhan Scheme:** (Galvanizing Organic Bio-Agro Resources Dhan) Focuses on managing cattle waste and agricultural residue to produce biogas and organic manure, which directly supports the goals of PM-PRANAM by providing natural fertilizer alternatives.
- **PKVY (Paramparagat Krishi Vikas Yojana):** A traditional farming improvement program launched in 2015 that provides financial assistance to farmers specifically for adopting organic farming practices and securing organic certifications.
- **Nano Urea and Nano DAP:** Developed indigenously by IFFCO, these are liquid nanotechnology-based fertilizers. A single 500ml bottle of Nano Urea can replace a conventional 45kg bag of granular urea, drastically reducing the bulk use of chemical fertilizers and minimizing soil and groundwater pollution.

- **MOVCDNER:** (Mission Organic Value Chain Development for North Eastern Region) A dedicated sub-mission to develop certified organic production in the northeastern states, linking growers with consumers to create a robust organic value chain.

Q 87. Ans. D

Explanation:

- India has set a target to achieve net-zero carbon emissions by the year **2070**. This historic commitment was officially announced by Prime Minister Narendra Modi during the 26th UN Climate Change Conference of the Parties (COP26) held in Glasgow in November 2021, as the final goal of India's five-part climate pledge known as the Panchamrit (five elixirs).
- Achieving "net-zero" means that the amount of greenhouse gases emitted into the atmosphere by a country is perfectly balanced by the amount removed or absorbed (through natural carbon sinks like forests or advanced carbon-capture technologies).
- **Additional Facts:**
 - ✓ As a stepping stone toward the 2070 net-zero goal, the Indian Cabinet recently approved updated Nationally Determined Contributions (NDCs) for 2031–2035. India has now committed to reducing the emissions intensity of its GDP by **47% by 2035** (compared to 2005 levels) and achieving **60%** of its cumulative electric power installed capacity from **non-fossil fuel** resources by **2035**.
 - ✓ **Panchamrit Strategy:** The original COP26 pledge included short-term 2030 targets to support the 2070 goal, including: reaching 500 GW of non-fossil energy capacity, meeting 50% of the country's energy requirements from renewable energy, and reducing total projected carbon emissions by one billion tonnes from 2021 to 2030.
 - ✓ India's 2070 target allows crucial carbon space to grow its economy and lift millions out of poverty, whereas developed economies like the US and the UK have set their net-zero target for 2050, and China has set its target for 2060.

Q 88. Ans. A

Explanation:

- **Statement 1 is correct:** Launched by the Bihar Police, "Operation Prahar" is an intensive state-wide drive aimed at the swift identification and immediate

arrest of individuals involved in serious criminal offenses (such as murder, attempt to murder, and major thefts) as well as continuous violators of the state's liquor prohibition laws.

- **Statement 2 is incorrect:** The operation is strictly a law enforcement initiative executed by the **Bihar Police**, not the disaster management authority. To effectively carry out these targeted raids and arrests, the police department constituted dedicated, quick-response units known as '**Vajra**' companies across all districts.

Q 89. Ans. A

Explanation:

- **Statement 1 is correct:** The PM MITRA (Pradhan Mantri Mega Integrated Textile Region and Apparel) scheme is designed to establish seven massive, world-class integrated textile parks across the country. These parks aim to integrate the entire textile value chain, from spinning, weaving, and processing to garment manufacturing, at single, large-scale locations.
- **Statement 2 is correct:** A core objective of the scheme is to attract Foreign Direct Investment (FDI), boost domestic manufacturing scale, reduce logistics costs, and ultimately position India as a highly competitive global hub for textile manufacturing and exports.
- **Statement 3 is incorrect:** The seven parks are not located exclusively in coastal states. The finalized sites include landlocked states as well. The seven locations are: Dhar (Madhya Pradesh), Lucknow (Uttar Pradesh), Warangal (Telangana), Amravati (Maharashtra), Navsari (Gujarat), Kalaburagi (Karnataka), and Virudhunagar (Tamil Nadu).
- **Additional Facts:**
 - ✓ **The 5F Vision:** PM MITRA is directly inspired by the Prime Minister's '5F' vision: Farm to Fibre to Factory to Fashion to Foreign. This emphasizes a seamless, traceable supply chain from agricultural cotton production all the way to global fashion exports.
 - ✓ **Plug-and-Play Infrastructure:** These parks are designed on a "plug-and-play" model, meaning core infrastructure (like reliable power, water supply, common effluent treatment plants with Zero Liquid Discharge, and workers' hostels) is pre-built. This allows businesses to set up factories immediately without worrying about baseline industrial development.
 - ✓ **National Technical Textiles Mission (NTTM):** While PM MITRA focuses on traditional and

integrated textiles, NTTM (launched with a ₹1,480 crore outlay) focuses specifically on high-tech “technical textiles” used in medical devices, defense, automotive, and geo-textile sectors.

- ✓ **Samarth Scheme:** To ensure these mega parks have the required skilled workforce, the Samarth scheme (Scheme for Capacity Building in Textiles Sector) runs in parallel to provide placement-oriented skilling across the organized textile value chain.

Q 90. Ans. A

Explanation:

- **CHAMPIONS Portal (a - 4):** Created by the **Ministry of MSME**, this platform stands for Creation and Harmonious Application of Modern Processes for Increasing the Output and National Strength. It serves as a unified control room and grievance redressal system to handhold smaller businesses.
- **MCA21 Version 3.0 (b - 1):** Launched by the **Ministry of Corporate Affairs**, this is India’s pioneer mission-mode e-governance initiative. The upgraded Version 3.0 heavily incorporates AI and machine learning to simplify corporate filings, compliance enforcement, and registry lookups.
- **Nivesh Mitra (c - 2):** Developed by the **Government of Uttar Pradesh**, it is an advanced single-window clearance system designed to enhance the ease of doing business by consolidating hundreds of industrial approvals and clearances under a single digital roof.
- **E-Shram Portal (d - 3):** Rolled out by the **Ministry of Labour and Employment**, it is a comprehensive national database aimed at registering unorganised workers (such as construction workers, gig workers, and street vendors) to streamline the targeted delivery of social security benefits.
- **Additional Facts:**
 - ✓ **Udyam Portal:** Operated by the **Ministry of MSME** as the official, completely paperless, and self-declaration-based portal for registering micro, small, and medium enterprises.
 - ✓ **SPICE+ (Simplified Proforma for Incorporating Company Electronically Plus):** Administered by the **Ministry of Corporate Affairs**, this single window web-form combines 10 distinct services across multiple ministries and departments to dramatically accelerate the process of company incorporation.
 - ✓ **Shram Suvidha Portal:** Maintained by the **Ministry of Labour and Employment** to

allocate a unique Labour Identification Number (LIN) to businesses, enabling the online submission of a single unified annual return for labor law compliance.

- ✓ **SAATHI Portal (Seed Traceability, Authentication and Holistic Inventory):** Developed by the **Ministry of Agriculture and Farmers Welfare** to tackle the challenge of spurious seeds through an end-to-end tracking system using QR-code tracing.
- ✓ **PRAGATI (Pro-Active Governance and Timely Implementation):** A unique, three-tier interactive platform monitored directly by the **Prime Minister’s Office (PMO)** that brings the PM, Union Secretaries, and State Chief Secretaries together to unblock delayed infrastructure projects.

Q 91. Ans. A

Explanation:

- **Garuda Shakti (a - 2):** A bilateral joint special forces exercise between India and **Indonesia**, aimed at enhancing military cooperation and interoperability in counter-terrorism and special operations.
- **Varuna (b - 1):** A bilateral naval exercise between the Indian Navy and the **French Navy**. (Note: India and France also conduct Garuda for the Air Force and Shakti for the Army).
- **Nomadic Elephant (c - 4):** An annual bilateral military exercise between the Indian Army and the **Mongolian Armed Forces**, focused on counter-insurgency and peacekeeping operations under the UN mandate.
- **Dharma Guardian (d - 3):** An annual joint military exercise between the Indian Army and the **Japanese Ground Self-Defense Force**, focusing on tactical drills and urban warfare.

Partner Nation	Army Exercise	Naval Exercise	Air Force Exercise
USA	Yudh Abhyas, Vajra Prahar	SALVEX	Cope India
United Kingdom	Ajeya Warrior	Konkan	Indradhanush
Sri Lanka	Mitra Shakti	SLINEX	-
Singapore	Bold Kurukshetra	SIMBEX	-
Nepal	Surya Kiran	-	-
Bangladesh	Sampriti	Bongosagar	-

Oman	Al Nagah	Naseem-al-Bahr	Eastern Bridge
Uzbekistan	Dustlik	-	-
Malaysia	Harimau Shakti	-	Udarashakti
Maldives	Ekuverin	-	-
Egypt	Cyclone	-	-

Q 92. Ans. B

Explanation:

- **Tandur Red Gram (a - 2):** A local variety of pigeon pea (tur dal) from **Telangana**, highly sought after for its excellent taste, better cooking quality, and high protein content.
- **Marcha Rice (b - 3):** Cultivated primarily in the West Champaran district of **Bihar**, it is famous for its distinct aroma, palatability, and black-pepper-like shape (hence the name 'Mircha' or 'Marcha').
- **Gamosa (c - 1):** A deeply revered traditional handwoven textile from **Assam**, characterized by its white body and red borders. It is a symbol of Assamese culture, respect, and identity.
- **Kangra Tea (d - 4):** Cultivated in the Kangra valley of **Himachal Pradesh**, this tea is renowned for its unique color and flavor. It recently gained international prominence by receiving a European Geographical Indication (PGI) tag.

State / UT	Notable GI Tags
Bihar	Mithila Makhana, Shahi Litchi, Magahi Paan, Zardalu Mango, Katarni Rice, Bhagalpuri Silk
Uttar Pradesh	Banarasi Paan, Ramnagar Bhanta (Brinjal), Adamchini Rice, Hathras Hing, Aligarh Taala
Jammu & Kashmir	Basohli Painting, Kashmiri Saffron, Gucchi Mushroom, Bhaderwah Rajmash
Ladakh	Raktsey Karpo (Apricot), Ladakh Wood Carving, Pashmina Wool
Madhya Pradesh	Sundaraja Mango (Rewa), Gond Painting, Kadaknath Chicken (Jhabua)
Chhattisgarh	Nagri Dubraj Rice, Jeeraphool Rice
Tamil Nadu	Cumbum Panneer Thrachai (Grapes), Kancheepuram Silk, Authoor Vetrilai (Betel leaf)
Assam	Kaji Nemu (Lemon), Chokuwa Rice, Judima (Rice Wine)

Q 93. Ans. B

Explanation:

- **V.O. Chidambaranar (VOC) Port**, located in

Thoothukudi, Tamil Nadu, has officially become the first port in India to successfully produce green hydrogen. India's first port-based Green Hydrogen Pilot Project was inaugurated here by the Ministry of Ports, Shipping, and Waterways.

- The pilot facility is designed to generate green hydrogen using renewable energy (electrolysis) to power streetlights and electric vehicle (EV) charging stations within the port colony, marking a significant milestone in India's maritime decarbonisation journey.

Additional Facts:

- ✓ **Deendayal Port (formerly Kandla, Gujarat):** Known as a tidal port, it holds the distinction of being the first Export Processing Zone (EPZ) established in India and Asia (set up in 1965).
- ✓ **Visakhapatnam Port (Andhra Pradesh):** It is India's deepest landlocked and naturally protected port. It is uniquely shielded from cyclones and ocean currents by a solid rock promontory known as the "Dolphin's Nose".
- ✓ **Paradip Port (Odisha):** It is the first major port commissioned on the East Coast of India after independence, strategically located at the confluence of the Mahanadi River and the Bay of Bengal.
- ✓ **Other VOC Port Records:** Apart from green hydrogen production, VOC Port recently became the first Indian port to deploy an advanced radar-based anti-drone security system. It also boasts the highest rooftop solar capacity among all major Indian ports.
- ✓ **Cochin Port (Kerala):** Situated on the man-made Willingdon Island, it is a magnificent natural harbour famously known as the "Queen of the Arabian Sea".

Q 94. Ans. D

Explanation:

- **Statement 1 is correct:** The 35th NATO Summit was held in June 2025 at The Hague, Netherlands. It marked the first time the Netherlands hosted the NATO summit, focusing heavily on increasing defence investment, strengthening Europe's defence industrial capacity, and sustained support for Ukraine.
- **Statement 2 is correct:** The 51st G7 Summit was hosted by Canada in June 2025 at the Kananaskis Resort in Alberta. The summit's core discussions revolved around economic security, critical minerals, and the governance of rapidly evolving technologies like AI and quantum computing.

- **Statement 3 is correct:** The 3rd UN Ocean Conference (UNOC3) took place in June 2025 in Nice, France. Co-hosted by the governments of France and Costa Rica, the conference aimed to mobilize global stakeholders to accelerate actions for the conservation and sustainable use of the oceans (aligning with Sustainable Development Goal 14: Life Below Water).

- **Additional Facts:**

- ✓ **20th G20 Summit:** Scheduled for late 2025 in Johannesburg, South Africa. This is a historic milestone as it marks the first time a G20 leaders' summit is hosted on the African continent. (The USA will host the G21 in 2026).
- ✓ **COP30 (UN Climate Change Conference):** Scheduled for November 2025 in Belém, Brazil. It is expected to be a highly critical summit as nations are required to submit their updated Nationally Determined Contributions (NDCs) for climate action.
- ✓ **AI Action Summit:** Hosted in Paris, France (February 2025). This summit built upon the foundational AI Safety Summit held in Bletchley Park (UK) to formulate concrete global AI governance and safety frameworks.
- ✓ **47th ASEAN Summit:** Hosted in Kuala Lumpur by Malaysia, which holds the 2025 ASEAN Chairmanship.
- ✓ **BRICS Summit 2025:** Hosted by Brazil, taking over the presidency from Russia (who hosted the 2024 summit in Kazan).

Exercise Name	Type / Participating Nations	Key Details
Malabar	Multilateral (India, USA, Japan, Australia)	The premier naval exercise of the Quad nations, crucial for maintaining a free and open Indo-Pacific.
MILAN	Multilateral (Hosted by India)	A biennial mega-naval exercise hosted by the Indian Navy, bringing together navies from over 50 friendly foreign countries.
SITMEX	Trilateral (India, Singapore, Thailand)	An annual trilateral maritime exercise focused on strengthening mutual confidence in the Andaman Sea.
DOSTI	Trilateral (India, Maldives, Sri Lanka)	A biennial Coast Guard exercise (recently expanded to include Bangladesh as an observer) focused on maritime search and rescue.
Al-Mohed Al-Hindi	Bilateral (India, Saudi Arabia)	India's first-ever joint naval exercise with the Royal Saudi Navy, marking a milestone in Gulf defence ties.
Zayed Talwar	Bilateral (India, UAE)	A bilateral naval exercise aimed at enhancing maritime security and cooperation in the Arabian Sea and Gulf regions.
Kakadu	Multilateral (Hosted by Australia)	A premier international maritime training exercise hosted by the Royal Australian Navy, in which the Indian Navy regularly participates.

Q 95. Ans. C

Explanation:

- **AUSINDEX (a - 1):** A major biennial bilateral maritime exercise between the Indian Navy and the Royal Australian Navy (RAN), focused on enhancing anti-submarine warfare and interoperability in the Indo-Pacific region.
- **JAIMEX / JIMEX (b - 2):** Refers to the bilateral maritime exercises conducted between the Indian Navy and the Japan Maritime Self-Defense Force (JMSDF).
- **IN-RoKN Exercise (c - 3):** A bilateral maritime partnership exercise between the Indian Navy (IN) and the Republic of Korea Navy (RoKN), reflecting growing strategic and defence ties between New Delhi and Seoul.
- **Konkan Exercise (d - 4):** An annual bilateral naval exercise conducted since 2004 between the Indian Navy and the United Kingdom's Royal Navy to build maritime synergy and share best practices.

Q 96. Ans. B

Explanation:

- **Pair 1 is incorrect:** The Namo Laxmi Scheme was launched by the Gujarat Government, not Maharashtra. It provides financial assistance (totaling ₹50,000 over four years) to school-going girls from Classes 9 to 12 to reduce dropout rates and promote adolescent health.
- **Pair 2 is correct:** The Mukhyamantri Mahila Rojgar Yojana was launched by the Bihar Government in September 2025. It aims to foster self-employment

and economic independence for women by offering an initial seed capital of ₹10,000, with potential additional support of up to ₹2 lakh for scaling micro-businesses.

- **Pair 3 is correct:** The **Pandit Deendayal Lado Laxmi Yojana** is a welfare initiative recently launched by the **Haryana Government**. Under this scheme, eligible women from low-income families receive ₹2,100 per month via Direct Benefit Transfer (DBT) to support their financial independence and well-being.

- **Additional Facts:**

- ✓ **Mukhyamantri Majhi Ladki Bahin Yojana (Maharashtra):** Provides ₹1,500 per month to eligible women aged 21 to 65 years to ensure financial independence and nutritional security.
- ✓ **Subhadra Yojana (Odisha):** A massive women's empowerment scheme providing ₹10,000 annually (in two equal installments) to over one crore women over five years.
- ✓ **Mahtari Vandana Yojana (Chhattisgarh):** Offers ₹1,000 per month to married women to improve their economic status and family decision-making power.
- ✓ **Mukhyamantri Maiyan Samman Yojana (Jharkhand):** Provides ₹1,000 per month to eligible women from economically weaker sections.

Q 97. Ans. B

Explanation:

- **Indore** (Madhya Pradesh) and **Udaipur** (Rajasthan) have officially become the first two Indian cities to be recognized as accredited "Wetland Cities" under the Ramsar Convention on Wetlands. While the Ministry of Environment, Forest and Climate Change had initially nominated three cities, Indore, Udaipur, and Bhopal, but only Indore and Udaipur successfully secured the accreditation in the initial cycle.
- **Indore:** Earned the recognition largely due to its exceptional conservation of the **Sirpur Lake** (a designated Ramsar site) and **Yashwant Sagar**, integrating waste management with ecological preservation to develop a thriving bird sanctuary.
- **Udaipur:** Popularly known as the 'City of Lakes', Udaipur was accredited for the sustainable urban management of its five major surrounding wetlands, **Lake Pichola**, **Fateh Sagar**, **Rang Sagar**, **Swaroop Sagar**, and **Doodh Talai**, which are vital to the city's microclimate and cultural identity.

- **Additional Facts:**

- ✓ **Wetland City Accreditation (WCA):** This voluntary accreditation system was approved in 2015 during COP12 of the Ramsar Convention. It formally recognizes cities that have taken exceptional steps to protect their urban and peri-urban wetlands.
- ✓ **Bhopal's Status:** Bhopal was the third city nominated by India. However, its accreditation was reportedly held back due to ecological concerns regarding a proposed road infrastructure project threatening the **Bhoj Wetland**.
- ✓ **The Ramsar Convention:** Signed in 1971 in Ramsar, Iran, it is a global intergovernmental treaty for the conservation and wise use of wetlands.
- ✓ **India's Ramsar Footprint:** As of recent updates, India holds the highest number of Ramsar-designated wetlands in South Asia. **Tamil Nadu** currently hosts the highest number of Ramsar sites among Indian states, followed by **Uttar Pradesh**.
- ✓ **Montreux Record:** A register under the Ramsar Convention that highlights wetland sites where severe ecological character changes have occurred or are occurring. Currently, two Indian sites are on this record: **Keoladeo National Park** (Rajasthan) and **Loktak Lake** (Manipur).

Q 98. Ans. C

Explanation:

- **Sonam Wangchuk** was featured in the **TIME100 Climate 2025** list, specifically under the 'Defenders' category. TIME recognized him for combining ancient cultural traditions with modern science to build "Ice Stupas", massive, artificial cone-shaped glaciers that store winter meltwater. These stupas act as a critical reservoir, releasing water to support local farmers during the dry, high-altitude spring months.
- **Additional Facts:**
 - ✓ **Categories of TIME100 Climate:** The list recognizes leaders driving tangible climate action globally across specific categories: **Titans**, **Catalysts**, **Innovators**, **Leaders**, and **Defenders**.
 - ✓ **The Ramon Magsaysay Award:** In 2018, Wangchuk received this prestigious honor (often referred to as Asia's Nobel Prize) for his community-driven reform of the learning

system in Ladakh, which drastically improved the academic outcomes of local youth.

- ✓ **HIAL:** Building on the success of SECMOL, he founded the Himalayan Institute of Alternatives, Ladakh (HIAL), a unique university dedicated to experiential learning and tackling the specific ecological and developmental challenges of mountain communities.
- ✓ **Chewang Norphel:** Often called the “Ice Man of India,” he is another legendary Ladakhi civil engineer who actually pioneered the initial concept of flat artificial glaciers before Wangchuk innovated the conical “Ice Stupa” design.

Q 99. Ans. A

Explanation:

- India's first **Semiconductor Innovation Museum** was inaugurated in **Hyderabad** (Telangana) in October 2025. The initiative was spearheaded by the Technology Chip Innovation Program (T-Chip) to showcase path-breaking research and technological advancements in the chip-making sector.
- The museum aims to position Hyderabad as a premier hub for semiconductor innovation by hosting monthly demo days, global showcases, and a unique 30-day “Innovation Residency” model where startups and global innovators can display their work to investors and the public. Key exhibits include India's first indigenous AI chip, AI-powered humanoid robots, and next-generation EV systems.
- **Additional Facts:**
 - ✓ **India Semiconductor Mission (ISM):** A specialised business division within the Digital India Corporation that serves as the nodal agency for driving semiconductor and display manufacturing initiatives across the country.
 - ✓ **Dholera Fab:** Tata Electronics (in partnership with Taiwan's Powerchip Semiconductor Manufacturing Corp) is setting up India's first commercial semiconductor fabrication facility (Fab) in Dholera, Gujarat.
 - ✓ **Sanand ATMP:** US-based semiconductor giant Micron Technology is establishing a massive Assembly, Test, Marking, and Packaging (ATMP) facility in Sanand, Gujarat.
 - ✓ **Morigaon Facility:** Tata Semiconductor Assembly and Test Pvt Ltd (TSAT) is building another major ATMP unit in Morigaon, Assam, which is a crucial step toward developing a high-tech ecosystem in the Northeast.

Q 100. Ans. C

Explanation:

- **World's first generational tobacco ban (a - 2):** In May 2025, the **Maldives** passed a landmark law prohibiting anyone born on or after January 1, 2007, from ever purchasing, using, or being sold tobacco and vaping products. It is effectively the first country in the world to implement a true, nationwide generational smoking ban.
- **World's first fully cashless society (b - 1):** Sweden achieved this historic milestone in 2025, transitioning almost entirely to mobile payments (driven heavily by the national app “Swish”), cards, and the central bank's pilot digital currency, the e-krona.
- **World's first AI generated genome virus (c - 3):** Researchers at Stanford University in the **USA** successfully designed the first fully functional viral genome using an AI language model called “Evo.” This AI-generated bacteriophage was successfully designed to target and kill antibiotic-resistant E. coli.
- **World's first permanent nuclear waste repository (d - 4):** Located in Eurajoki, **Finland**, the Onkalo facility is built deep inside 1.9-billion-year-old bedrock. It is designed to safely store spent nuclear fuel for 100,000 years using an advanced multi-barrier copper and clay isolation system.
- **Additional Facts:**
 - ✓ **First 3D Wind Data Satellite Network:** A Hong Kong startup recently announced plans to launch the world's first satellite constellation designed specifically to optimize 3D wind data for weather prediction and renewable energy optimization.
 - ✓ **First Wooden Satellite (LignoSat):** Developed by Japanese researchers at Kyoto University, it was launched to test sustainable, biodegradable materials in space. The goal is to use wood that completely burns up upon re-entry, preventing the accumulation of harmful aluminum ash in the upper atmosphere.
 - ✓ **World's First Hydrogen-powered Passenger Train Network:** Germany pioneered the Coradia iLint, a regional train that operates entirely on hydrogen fuel cells, emitting only steam and condensed water as exhaust.

Q 101. Ans. D

Explanation:

- **Statement 1 is correct:** The inaugural (first-ever) edition of the Khelo India Tribal Games (KITG)

was held from March 25 to April 3, 2026, and was successfully hosted by the state of Chhattisgarh.

- **Statement 2 is correct:** The multi-sport event was organized across three key cities in the state to maximize regional participation: **Raipur, Jagdalpur (Bastar region), and Surguja.**
- **Statement 3 is correct:** **Karnataka** emerged as the overall champion, securing the first position in the final medal tally with 23 gold, 8 silver, and 7 bronze medals. Odisha finished second (crossing a half-century of total medals), while Jharkhand secured the third spot. The host state, Chhattisgarh, finished ninth.
- **Additional Facts:**
 - ✓ **Mascot:** The official mascot for the games was “**Morveer**”, a name rooted in the Chhattisgarhi dialect where “Mor” means “our own” and “Veer” symbolizes bravery and strength.
 - ✓ **Sports Featured:** The event featured seven competitive medal sports: Archery, Athletics, Football, Hockey, Swimming, Weightlifting, and Wrestling. Additionally, traditional indigenous sports like **Mallakhamb** and **Kabaddi** were included as special demonstration events.
 - ✓ **Outstanding Athletes:** Karnataka’s swimmer Manikanta L was crowned the most successful athlete of the Games, bagging a massive eight gold medals and one silver. In the women’s category, Odisha’s Anjali Munda clinched five gold medals in swimming.
 - ✓ **Strategic Objective:** Jointly organized by the Ministry of Youth Affairs and Sports, SAI, and the IOA, this exclusive tournament aims to bring structured sporting opportunities directly to tribal-majority regions (like the Dandakaranya belt), ensuring grassroots talent is identified, nurtured, and elevated to a national platform.

Q 102. Ans. C

Explanation:

- **Naxal-free India (2026) is correctly matched:** The Union Home Minister has set a strict, targeted deadline to completely eliminate Left Wing Extremism (Naxalism) from the country by March 31, 2026.
- **Confirmed railway tickets for all passengers (2027) is correctly matched:** Indian Railways has outlined a massive capacity expansion plan, which includes laying 4,000–5,000 km of new tracks annually and introducing 3,000 new trains, with the ultimate

goal of eliminating waiting lists and providing confirmed tickets to every passenger by 2027.

- **TB-free India (2030) is incorrectly matched:** While 2030 is the global target set by the UN’s Sustainable Development Goals (SDGs), the Government of India has set a much more ambitious domestic target to achieve a TB-free India by **2025** (five years ahead of the global timeline) through the Pradhan Mantri TB Mukh Bharat Abhiyaan.
- **\$1 trillion Digital Economy (2028) is correctly matched:** Originally envisioned for 2025–2026, the Ministry of Electronics and Information Technology (MeitY) recently updated the target year to achieve a \$1 trillion digital economy to the financial year 2027–2028 (FY28), accounting for post-pandemic adjustments.
- **Additional Facts (Other Major Government Target Years):**
 - ✓ **2025:** Blending 20% ethanol in petrol (E20 target, advanced from 2030).
 - ✓ **2027:** India projected to become the world’s third-largest economy (surpassing Japan and Germany).
 - ✓ **2030:** Achieving 500 GW of installed non-fossil energy capacity and generating 50% of the country’s total energy requirements from renewable sources.
 - ✓ **2030:** Eradicating Malaria from the country.
 - ✓ **2030:** Achieving 30% electric vehicle (EV) penetration in private cars, 70% in commercial vehicles, and 80% in two and three-wheelers.
 - ✓ **2047:** Achieving “Energy Independence” and becoming a fully developed nation (Viksit Bharat).
 - ✓ **2070:** Achieving Net-Zero carbon emissions.

Q 103. Ans. B

Explanation:

- Following India’s Operation Sindoor, launched on 7 May 2025 as a precision tri-service strike targeting terrorist infrastructure in Pakistan and Pakistan-occupied Kashmir (PoK) in response to the 22 April 2025 Pahalgam terror attack, Pakistan launched its retaliatory military operation on 10 May 2025, codenamed Operation Bunyan-un-Marsoos (also transliterated as Bunyan al-Marsus or Bunyanum Marsoos).
- **The name derives from a Quranic verse (Surah As-Saff, 61:4)** — “Surely Allah loves those who fight in His cause in solid ranks, as if they were one concrete/cemented structure” — chosen after this

phrase was reportedly found inscribed on a wall at a mosque/religious complex in Muzaffarabad that had been struck during Operation Sindoor.

● **Additional facts:**

- ✓ **Operation Swift Retort** : Pakistan's actual codename for its retaliatory strikes after India's 2019 Balakot air strikes — a different, earlier conflict.
- ✓ **Operation Zarb-e-Azb** : Pakistan's 2014 counter-terrorism military operation against militants in North Waziristan — unrelated to the India conflict.
- ✓ **Operation Radd-ul-Fasaad** : Pakistan's 2017 nationwide counter-terrorism operation — also unrelated to the 2025 India-Pakistan conflict.

Q 104. Ans. A

Explanation:

- Iran's parliament began drafting legislation to withdraw from the Nuclear Non-Proliferation Treaty (NPT) starting 16–18 June 2025, in the aftermath of the "Twelve-Day War" between Iran and Israel. This move followed:
- The IAEA Board of Governors' resolution, which for the first time in nearly two decades formally declared Iran in breach of its NPT safeguards obligations, citing lack of cooperation with inspections and unexplained uranium traces at undeclared sites.
- Israel's military campaign ("Operation Rising Lion"), beginning 13 June 2025, which struck Iranian nuclear facilities, military command centers, and killed several nuclear scientists — justified by Israel as necessary to prevent Iran from developing nuclear weapons.
- In response, Iran's Foreign Ministry confirmed parliament was preparing a bill on NPT withdrawal, though officials maintained Iran's stated policy still opposes developing nuclear weapons.

Q 105. Ans. B

Explanation:

- Prime Minister Narendra Modi was conferred the title "Dharma Chakravarti" on 28 June 2025 at Vigyan Bhawan, New Delhi, during the centenary celebrations of Jain saint Acharya Shri 108 Vidyanand Ji Maharaj.
- The event marked the formal launch of a year-long national tribute (28 June 2025 – 22 April 2026) to Acharya Vidyanand Ji, organised by the Union

Ministry of Culture in collaboration with the Bhagwan Mahaveer Ahimsa Bharti Trust, Delhi.

- The date itself was significant, as it marked the anniversary of the day (28 June 1987) that Acharya Vidyanand Ji himself had received the title of "Acharya."
- Accepting the honour, PM Modi remarked that he did not consider himself worthy of it, but accepted it humbly as "prasad" from the saints and dedicated it to "Maa Bharati."
- PM Modi, along with Union Culture Minister Gajendra Singh Shekhawat, also unveiled commemorative postage stamps in memory of the Acharya during the event.

Q 106. Ans. C

Explanation:

- Sub Lieutenant Aastha Poonia became the first woman fighter pilot of the Indian Navy on 3 July 2025, when she was awarded the coveted "Wings of Gold" by Rear Admiral Janak Bevli, Assistant Chief of Naval Staff (Air), at INS Dega, Visakhapatnam — during the graduation ceremony of the Second Basic Hawk Conversion Course (alongside Lieutenant Atul Kumar Dhull).
- Her induction into the Navy's fighter stream — until then an exclusively male domain — marks a historic breakthrough distinct from earlier roles held by women in Naval aviation (helicopters, maritime reconnaissance, and transport aviation).

Q 107. Ans. D

Explanation:

- The Bihar Cabinet, chaired by Chief Minister Nitish Kumar, approved a domicile-based policy on 8 July 2025 for the existing 35% horizontal reservation for women in state government jobs — one of 43 proposals cleared in that cabinet meeting (which also included the formation of a Bihar Youth Commission).
- Bihar has provided 35% reservation for women in government jobs since 2016. Until this change, women from any state were eligible to claim this quota if they secured a Bihar government post.

Q 108. Ans. C

Explanation:

- As per the National Crime Records Bureau (NCRB) 'Crime in India 2024' report, Kolkata has once again recorded the lowest crime rate (83.9 cases per lakh population) among major metropolitan cities,

maintaining its status as the safest metro city in the country. In contrast, Delhi recorded the highest crime rate among the major metros.

Q 109. Ans. C

Explanation:

- **Operation RAGEPILL:** Launched by the Narcotics Control Bureau (NCB), this intelligence-driven operation targeted a major international drug smuggling network.
- **What was seized:** The NCB achieved India's first-ever interception of **Captagon** (a highly addictive synthetic amphetamine-type psychotropic stimulant), confiscating 227.7 kilograms of the drug with an estimated international market value of ₹182 crore.
- Often referred to as the "Jihadi Drug" due to its widespread use by extremist groups (such as ISIS) in conflict zones to suppress fear, hunger, and fatigue, the contraband was found ingeniously hidden inside commercial chapati-cutting machines and sheep wool cargo. The syndicate was exploiting India as a transit corridor to traffic the drugs into Gulf countries like Saudi Arabia, leading to the arrest of a Syrian national who was operating out of New Delhi.
- **Additional Facts:**
 - ✓ **Operation MED MAX (2025):** The NCB dismantled a transnational pharmaceutical trafficking network spanning over 10 countries across four continents. It exposed an India-based network supplying controlled medicines (like tramadol) to the US, Europe, and Australia via encrypted platforms, drop-shipping models, and cryptocurrency.
 - ✓ **Operation Crystal Fortress (2025):** An NCB operation that resulted in the massive seizure of 328 kg of high-quality methamphetamine in Delhi.
 - ✓ **Operation Ketamelon (2025):** The NCB busted one of India's largest darknet drug syndicates, seizing substantial quantities of LSD blots, ketamine, and illicit cryptocurrency assets.
 - ✓ **Operation Chakravyuh (2026):** An intelligence-driven drive by the Directorate of Revenue Intelligence (DRI) that seized synthetic drugs like methaqualone and methamphetamine across multiple cities in Kerala. The drugs were intercepted across smuggling channels including airport passenger baggage and outbound export cargo.
 - ✓ **Operation Garuda:** A massive multi-phase

crackdown led by the Central Bureau of Investigation (CBI) in close coordination with the NCB, INTERPOL, and state police forces to dismantle international drug networks, maintaining a special strategic focus on the Indian Ocean Region.

Q 110. Ans. D

Explanation:

- The term "**ecocide**" literally translates to "killing the environment or Unlawful or wanton acts committed with knowledge that there is a substantial likelihood of severe, widespread, or long-term damage to the environment.
- The definition is the official legal definition drafted in 2021 by the Independent Expert Panel for the Legal Definition of Ecocide.
- There is currently a massive global campaign by environmental lawyers and climate activists to amend the Rome Statute of the International Criminal Court (ICC) to recognise "ecocide" as the **fifth international crime**, standing alongside Genocide, Crimes Against Humanity, War Crimes, and the Crime of Aggression. If accepted, it would allow corporate leaders and heads of state to be criminally prosecuted at the Hague for catastrophic environmental destruction (e.g., massive oil spills, Amazon deforestation, deep-sea mining disasters).
- **Additional Facts:**
 - ✓ **Urbicide:** The deliberate and systematic destruction of a city or its urban infrastructure. This term is heavily used in modern geopolitical analysis to describe the flattening of urban centers in modern warfare (e.g., in conflicts in Gaza, Ukraine, and Syria) to destroy the civilian capacity to live there.
 - ✓ **Domicide:** The massive, deliberate destruction of homes and residential infrastructure, essentially murdering the concept of "home" for a targeted population and causing mass displacement.
 - ✓ **Epistemicide:** The systematic destruction or erasure of an indigenous, traditional, or marginalized knowledge system, often carried out by a dominant colonial or cultural power.
 - ✓ **Biocide:** The destruction of life. While it can broadly mean the killing of living organisms, it is most frequently used in environmental literature to describe the mass application of toxic chemical pesticides and herbicides that poison entire local food webs.

- ✓ **Femicide (or Feminicide):** The intentional, targeted killing of women or girls specifically because of their gender, often driven by entrenched systemic misogyny or domestic violence.
- ✓ **Gendercide:** The systematic killing of members of a specific sex or gender (e.g., female infanticide or the targeted killing of military-aged males during a conflict).

Q 111. Ans. C

Explanation:

- Madagascar is the largest island in the Indian Ocean and the fourth-largest island in the world, covering an area of about 587,000 sq. km. It is located off the southeastern coast of Africa and is separated from the mainland by the Mozambique Channel. Madagascar is renowned for its exceptional biodiversity, with a high proportion of endemic plant and animal species.
- **Additional Facts:**
 - ✓ Sri Lanka – A large island south of India, but much smaller than Madagascar.
 - ✓ Maldives – An archipelago consisting of numerous coral islands, not a single large island.
 - ✓ Réunion – A volcanic island east of Madagascar, significantly smaller in area.

Q 112. Ans. C

Explanation:

- **Statement 1 is incorrect:** Neap tides occur during the first quarter and third (last) quarter phases of the Moon, when the gravitational forces of the Sun and Moon act at right angles to each other. This results in the least difference between high and low tides.
- **Statement 2 is incorrect:** Spring tides occur during the new moon and full moon, when the Sun, Earth, and Moon are aligned (syzygy). The combined gravitational pull of the Sun and Moon produces the greatest difference between high and low tides.
- **Additional Facts:**
 - ✓ Spring Tide = Sun, Earth, Moon in a Straight Line → New Moon & Full Moon → Highest tidal range.
 - ✓ Neap Tide = Right Angle between Sun and Moon → First & Third Quarter Moon → Lowest tidal range.

Q 113. 113. Ans. D

Explanation:

- **Statement 1 is incorrect:** Ox-bow lakes are

not formed by glacial erosion. They are fluvial landforms created when a river cuts through the narrow neck of a meander, leaving the abandoned loop as a crescent-shaped lake. Ox-bow lakes are also known as cut-off lakes and are commonly found in the floodplains of mature rivers such as the Ganga, Mississippi, and Brahmaputra.

- **Statement 2 is incorrect:** Ox-bow lakes are not depositional features of a river delta. They develop due to the combined action of lateral erosion and deposition during the meandering stage of a river. Deltaic features include distributaries, levees, and marshes, whereas ox-bow lakes are meander-related landforms.
- **Statement 3 is incorrect:** Ox-bow lakes are not characteristic of the upper course of a river. They are mainly found in the middle and lower courses, where rivers flow over gentle slopes and develop large meanders. The upper course is dominated by vertical erosion, producing landforms such as V-shaped valleys, gorges, rapids, and waterfalls.

Q 114. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Tropical rainforests are characterized by a dense, multi-layered evergreen canopy, consisting of the emergent layer, canopy, understory, and forest floor. The thick canopy intercepts most sunlight, resulting in rich biodiversity but relatively sparse ground vegetation. These forests occur in regions with high temperatures and annual rainfall exceeding 200 cm, such as the Amazon, Congo Basin, and Southeast Asia.
- **Pair 2 is correctly matched:** Temperate grasslands possess deep, humus-rich, and highly fertile soils, making them among the world's most productive agricultural regions. These soils are known as Chernozem in the Eurasian Steppes and Mollisols in the North American Prairies. The natural vegetation is dominated by grasses due to moderate rainfall and periodic droughts.
- **Pair 3 is incorrectly matched:** Permafrost is not a characteristic feature of deserts. Deserts are defined by very low annual rainfall (less than 25 cm), sparse xerophytic vegetation, and large diurnal temperature ranges. While cold deserts such as the Gobi exist, the presence of permafrost is not a defining feature of desert biomes.
- **Pair 4 is correctly matched:** The Tundra biome is characterized by permanently frozen subsoil (permafrost), which restricts root penetration and

prevents the growth of trees. Consequently, only mosses, lichens, sedges, grasses, and dwarf shrubs can survive. During the short summer, only the thin surface layer thaws, creating marshy conditions and supporting migratory birds.

Q 115. Ans. A

Explanation:

- **Statement 1 is correct:** In Stage 2 (Early Expanding Stage) of the Demographic Transition Theory, the death rate declines rapidly due to improvements in healthcare, sanitation, nutrition, clean drinking water, and disease control. As a result, life expectancy increases while infant and child mortality decline significantly.
- **Statement 2 is correct:** During this stage, the birth rate remains high because social customs, limited access to contraception, and agrarian lifestyles continue to encourage large families. The decline in birth rates usually begins only in the next stage (Stage 3).
- **Statement 3 is incorrect:** Population growth is not slow in Stage 2; rather, it is very rapid because birth rates remain high while death rates fall sharply, creating a large natural increase. This phase is often referred to as the “population explosion” stage, commonly experienced by many developing countries during the early phases of modernization.
- **Additional Fact:**
 - ✓ **Stage 1:** High birth rate + High death rate → Stable population
 - ✓ **Stage 2:** High birth rate + Rapidly declining death rate → Population explosion
 - ✓ **Stage 3:** Declining birth rate + Low death rate → Growth slows
 - ✓ **Stage 4:** Low birth rate + Low death rate → Stable population
 - ✓ **Stage 5 (proposed):** Very low birth rate + Low death rate → Population decline or ageing population

Q 116. Ans. B

Explanation:

- A circular settlement is one in which houses are arranged around a central feature such as a pond, village common, cattle enclosure, fort, or place of worship. This pattern provides better security, promotes community interaction, and allows easy access to the central resource. It is commonly found in regions where defence, water availability, or livestock rearing has historically influenced settlement planning.

● Additional Facts:

- ✓ **Star-shaped settlement:** Develops where roads radiate outward from a central point, giving the settlement a star-like appearance. It is commonly associated with transport junctions rather than a central communal feature.
- ✓ **T-shaped settlement:** Develops along a T-junction of roads, with houses concentrated along the intersecting routes. The road network is the dominant factor determining its layout.
- ✓ **Cross-shaped settlement:** Forms at the intersection of two major roads, with buildings extending along all four directions. It is primarily influenced by transportation routes and trade.

Q 117. Ans. A

Explanation:

- **a. Equatorial⁸ 4 (Dense broadleaf evergreen forests, Selvas):** The Equatorial climate, characterised by high, uniform temperatures and heavy rainfall throughout the year, supports dense, multi-layered broadleaf evergreen rainforests — locally known as Selvas in the Amazon basin, and similarly found in the Congo basin and Southeast Asia.
- **b. Mediterranean⁸ 2 (Sclerophyllous shrubs, e.g., Chaparral):** The Mediterranean climate, with its hot, dry summers and mild, wet winters, favours drought-resistant, hard-leaved (sclerophyllous) vegetation — called Chaparral in California, Maquis in the Mediterranean basin itself, Fynbos in South Africa, and Mallee scrub in Australia.
- **c. Tundra⁸ 1 (Mosses and lichens):** The Tundra climate, with extremely low temperatures and a very short growing season, supports only low-growing vegetation like mosses, lichens, and dwarf shrubs, since trees cannot survive the harsh cold and permafrost conditions.
- **d. Temperate Grassland⁸ 3 (Tall grasses and sparse trees, e.g., Steppes):** Temperate grasslands, found in continental interiors with moderate rainfall, support extensive grass cover with few trees — known regionally as Steppes (Eurasia), Prairies (North America), Pampas (South America), and Downs (Australia).

Q 118. Ans. B

Explanation:

- The Peru–Chile Trench (also called the Atacama Trench), one of the deepest oceanic trenches in the world, runs along the western coast of South

America. It has formed due to the subduction of the oceanic Nazca Plate beneath the continental South American Plate — a classic example of a convergent (destructive) plate boundary.

- As the denser oceanic Nazca Plate is forced downward beneath the lighter continental South American Plate, it creates this deep linear depression on the ocean floor.
- This same subduction process is directly responsible for the uplift of the Andes Mountains along the western margin of South America, as compressive forces from the colliding plates fold and thicken the continental crust.
- It also explains the region's intense seismic and volcanic activity — the Peru-Chile Trench lies along the eastern rim of the Pacific Ring of Fire, making this one of the most earthquake- and volcano-prone zones on Earth (e.g., the 1960 Valdivia earthquake in Chile, the largest ever recorded).

Q 119. Ans. D

Explanation:

- **The major factors that influence the distribution of temperature across the Earth's surface, as per standard physical geography, include:**
 - ✓ **Latitude:** The angle of incidence of the sun's rays varies with latitude — areas near the Equator receive more direct, concentrated solar radiation, while areas near the poles receive slanting, dispersed rays, resulting in a general temperature decrease from the Equator toward the poles.
 - ✓ **Altitude:** Temperature decreases with increasing height above sea level (the normal lapse rate), making mountainous regions cooler than nearby lowlands at the same latitude.
 - ✓ **Distance from the sea / Distribution of land and water:** Land heats up and cools down faster than water (differential heating), so continental interiors experience greater temperature extremes (continentality), while coastal areas have more moderate, equable temperatures due to the moderating influence of the sea.
 - ✓ **Ocean currents:** Warm currents (like the Gulf Stream) raise temperatures of coastal areas they pass, while cold currents (like the Peru/Humboldt Current) lower them, significantly influencing regional climates.
- **Other factors:** Prevailing winds, aspect/slope of land, cloud cover, and vegetation cover also play a role.

Q 120. Ans. A

Explanation:

- The Duncan Passage is a strait in the Bay of Bengal, roughly 48 km wide, that separates Rutland Island (part of the Great Andaman/South Andaman group) to the north from Little Andaman Island to the south. The Bay of Bengal lies to its west, and the Andaman Sea to its east.

Q 121. Ans. B

Explanation:

- **Assertion (A) is true:** Rajendra Chola I (r. 1014–1044 CE) launched a large-scale naval expedition against the Srivijaya kingdom (in present-day Sumatra and the Malay Peninsula) around 1025 CE. This was undertaken to protect and promote Chola trading interests in Southeast Asia and to break Srivijaya's monopoly over the lucrative maritime trade routes between India and China, particularly the Malacca Strait. The Cholas captured several Srivijayan ports, including Kadaram (Kedah) and Palembang.
- **Reason (R) is true:** Rajendra Chola I did found the new capital city of Gangaikonda Cholapuram, commemorating his victorious northern military campaign that reached the banks of the Ganga (Ganges) — hence the name “Gangaikonda Cholapuram,” meaning “the city of the Chola who took the Ganga.”
- **Why (R) is not the correct explanation of (A):**
 - ✓ Both statements describe genuine, significant military achievements of Rajendra Chola I, but they are two entirely separate campaigns in different directions — the northern campaign (against the Pala kingdom of Bengal, reaching the Ganga) and the naval expedition against Srivijaya (a maritime campaign in Southeast Asia) — with no causal link between them.
 - ✓ The founding of Gangaikonda Cholapuram was a direct commemoration of the northern land campaign, not of the naval expedition against Srivijaya.

Q 122. Ans. B

Explanation:

- The Pagla Panthis Movement was founded by Karam Shah and later led by Tipu Shah in northern Bengal. It began as an egalitarian religious sect that attracted peasants, artisans, and lower social groups by emphasising equality and social justice. Over time, the movement acquired a strong agrarian character and turned into a peasant resistance

against the oppressive practices of zamindars and excessive revenue demands under British rule.

- The movement became particularly influential in the districts of Mymensingh and Sherpur and challenged both zamindari authority and colonial administration.

Q 123. Ans. D

Explanation:

- King Charles II of England received the islands of Bombay as part of the dowry when he married Catherine of Braganza, the Portuguese princess (daughter of King John IV of Portugal), in 1661. This marriage treaty transferred Bombay from Portuguese control to the English Crown.
- Charles II subsequently leased Bombay to the English East India Company in 1668 for a nominal annual rent of £10 in gold, since the Crown found it difficult and expensive to administer and defend the islands directly. This transfer proved hugely consequential — the Company developed Bombay into a major fortified trading settlement, and it eventually grew into one of British India's three principal Presidency towns (alongside Madras and Calcutta).

Q 124. Ans. C

Explanation:

- **a. Jai Singh of Amber** ⁸ 2 (Title 'Mirza Raja Sawai' and Governorship of Malwa): To secure Rajput loyalty, Jai Singh (Sawai Jai Singh II) was conferred the title "Mirza Raja Sawai" and appointed governor of Malwa, as part of the broader conciliatory policy toward Rajput chieftains.
- **b. Ajit Singh of Marwar** → 1 (Title 'Maharaja' and Governorship of Gujarat): Ajit Singh was granted the title of Maharaja and made governor of Gujarat, cementing Marwar's alliance with the Mughal court after years of Rajput rebellion following Aurangzeb's death.
- **c. King Shahu** → 4 (Granted both Chauth and Sardeshmukhi of the Deccan): Under the 1719 settlement (confirmed by the Sayyid Brothers, building on the earlier 1711 arrangement with Mughal deputy Daud Khan Panni), Shahu was granted Shivaji's swarajya along with the right to collect both Chauth and Sardeshmukhi across the six Deccan provinces, on condition that Mughal officials would make the collection and hand it to Maratha officers.
- **d. Maratha Sardars** → 3 (Granted only Sardeshmukhi of the Deccan): Unlike Shahu's comprehensive settlement, the wider body of Maratha sardars

operating under him were accommodated with a more limited concession — Sardeshmukhi rights alone — reflecting the Mughals' more cautious, half-hearted approach toward the broader Maratha nobility.

Q 125. Ans. A

Explanation:

- **Statement 1 is correct:** Sir Charles Napier officially annexed Sind in 1843, during the governor-generalship of Lord Ellenborough. The annexation followed a brief conflict in which Napier defeated the Baluchi Amirs of Sind at the Battles of Miani (Meeanee) and Hyderabad. Napier is famously (though possibly apocryphally) attributed with the one-word Latin pun report "Peccavi" ("I have sinned/Sind").
- **Statement 2 is incorrect:** The Treaty of Yandabo (1826) ended the First Anglo-Burmese War (1824–1826), not the First Anglo-Afghan War. Under this treaty, Burma ceded Assam, Manipur, Arakan, and Tenasserim to the British and paid a large war indemnity. The First Anglo-Afghan War (1838–1842) was a completely separate conflict, which ended in a catastrophic retreat and near-total destruction of the British/Company army during its withdrawal from Kabul in January 1842 — not with a treaty called "Yandabo." This statement conflates two unrelated wars (Burma and Afghanistan).

Q 126. Ans. A

Explanation:

- **Statement 1 is correct:** The Permanent Settlement, introduced by Lord Cornwallis in 1793 in Bengal, Bihar, and Orissa, fixed the state's share of land revenue permanently and in perpetuity with the zamindars, who became hereditary proprietors of the land in exchange for guaranteed revenue payments to the Company.
- **Statement 2 is correct:** The Ryotwari system, pioneered by Thomas Munro and Alexander Read and extended across much of the Madras and Bombay Presidencies, involved the government settling revenue directly with individual peasant cultivators (ryots) rather than through any intermediary — this bypassed not just zamindars, but also the collective, headman-mediated model used under the Mahalwari system, where village headmen (lambardars) were responsible for collecting and remitting revenue on behalf of the whole village community.
- **Statement 3 is incorrect:** The Mahalwari system, devised by Holt Mackenzie in 1822 (revised by Lord William Bentinck in 1833), was implemented

in the North-Western Provinces (present-day Uttar Pradesh), Punjab, parts of Central India, and the Gangetic Valley — not in Malabar. Malabar, part of the Madras Presidency in southern India, actually fell under the Ryotwari system, not Mahalwari.

Q 127. Ans. C

Explanation:

- **a. Cornwallis Code⁸ 1 (Separation of revenue administration from judicial functions):** Cornwallis's reforms (1793) separated the judiciary from revenue administration by removing judicial powers from the Collector and vesting them in a separate District Judge, ending the earlier practice of the same official both collecting revenue and adjudicating disputes.
- **b. High Courts Act, 1861⁸ 2 (Amalgamation of Supreme Courts and Sadar Diwani Adalats):** This Act provided for the creation of High Courts at Calcutta, Bombay, and Madras by amalgamating the Crown's Supreme Courts with the Company's Sadar Diwani and Sadar Nizamat Adalats, unifying two previously parallel judicial systems.
- **c. Aitchison Commission, 1886⁸ 3 (Recommended raising the maximum age for civil services examination):** The Public Service Commission under Sir Charles Aitchison recommended raising the ICS examination age limit back to 23 (after it had been progressively lowered to 19 by 1878), along with proposing a Statutory Civil Service to increase Indian representation.
- **d. Macaulay Committee, 1854⁸ 4 (Institutionalised open competitive examinations for the civil services):** Constituted under the provisions of the Charter Act of 1853, the Macaulay Committee recommended abolishing the Company's patronage-based nomination system entirely in favour of open competitive examinations, laying the foundation of the modern Indian Civil Service (implemented from 1855).

Q 128. Ans. A

Explanation:

- **Statement 1 is correct:** Lord Dalhousie annexed the kingdom of Awadh in February 1856 entirely on grounds of misgovernance/misrule under Nawab Wajid Ali Shah. This is notable precisely because Awadh's annexation did not rely on the Doctrine of Lapse (unlike Satara, Jhansi, or Nagpur) — since Awadh had a legitimate ruler with heirs — making it a distinct and separate justification, purely based

on alleged chronic maladministration, despite the Nawab's loyalty to the Company. This annexation is widely considered one of the key triggers of the Revolt of 1857, especially given widespread resentment in the Awadh region (home to a large section of the Bengal Army's sepoys).

- **Statement 2 is incorrect:** Lord Lytton was the Viceroy who enacted (not repealed) the Vernacular Press Act in 1878, nicknamed the "Gagging Act," to curb criticism of British policy in Indian-language newspapers — partly provoked by press criticism of his handling of the 1876–78 famine and the Second Anglo-Afghan War. It was his successor, Lord Ripon, who repealed the Act in 1882, as part of his broader liberal administrative reforms.

Q 129. Ans. A

Explanation:

- **Statement 1 is correct:** The Censorship of Press Act, 1799, was enacted by Lord Wellesley, driven by fears of a possible French invasion of India (given the ongoing Napoleonic Wars) and anxiety over press criticism during his aggressive expansionist policies. It imposed strict pre-censorship requirements on all newspapers.
- **Statement 2 is correct:** Sir Charles Metcalfe, acting Governor-General in 1835, repealed the restrictive press licensing regulations (the "Press Ordinance" of 1823 introduced by John Adam), earning him the honorary title "Liberator of the Indian Press." The Metcalfe Act of 1835 removed the requirement for prior government licensing and allowed the press significantly greater freedom of expression.
- **Statement 3 is incorrect:** The Vernacular Press Act, 1878, was widely condemned precisely because it denied publishers any right of judicial appeal. Decisions by district magistrates — including seizure of printing presses, demands for security deposits, and pre-publication censorship — were final and left entirely to executive discretion, with no provision for a judicial trial or appeal in the High Courts. This absence of due process was one of the Act's most criticized "draconian" features, earning it the nickname "the Gagging Act."

Q 130. Ans. C

Explanation:

- **The First Anglo-Sikh War (1845–46) ended with the Treaty of Lahore, signed on 9 March 1846, between the British East India Company and the Sikh Durbar (representing the young Maharaja Duleep Singh). Its key provisions included:**

- A massive war indemnity of over 1.5 crore rupees, which the depleted Sikh treasury could not fully pay.
- Cession of the Jalandhar Doab (the territory between the Beas and Sutlej rivers) to the British.
- Since the Sikhs could not pay the full indemnity, they ceded Kashmir and Hazara as compensation, which the British then separately sold to Raja Gulab Singh of Jammu for 75 lakh rupees under a supplementary agreement (the 1846 Treaty of Amritsar) — creating the princely state of Jammu & Kashmir.
- A drastic reduction of the Sikh (Khalsa) army.
- Stationing of a British Resident at Lahore, with British troops to remain temporarily.

Q 131. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Government of India Act, 1935, did propose an All-India Federation combining British Indian provinces with the princely states, but this federal scheme never actually came into force. It required accession by princely states representing at least half of their total population and half of the seats allotted to them in the proposed Federal Legislature a threshold that was never met, primarily due to the reluctance of princely rulers wary of ceding authority and losing autonomy. As a result, the federal part of the 1935 Act remained a dead letter throughout British rule.
- **Statement 2 is correct:** The Act abolished dyarchy at the provincial level — which had been introduced earlier under the Government of India Act, 1919 (Montagu-Chelmsford Reforms) and replaced it with full provincial autonomy, under which provinces were given separate legal identities with elected ministers responsible to provincial legislatures across all departments. However, the Act simultaneously introduced dyarchy at the Centre (federal level), dividing subjects into “reserved” (retained under the Governor-General’s control, e.g., defence, foreign affairs) and “transferred” (administered with the aid of ministers responsible to the legislature).

Q 132. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Lord Mayo, Viceroy from 1869–1872, commissioned the Statistical Survey of India in 1869 under Sir William Wilson Hunter, who produced detailed district-wise gazetteers covering the economic, social, and demographic conditions of British India — a foundational effort in colonial data-driven governance.

- **Pair 2 is correctly matched:** Lord Curzon (Viceroy 1899–1905) passed the Ancient Monuments Preservation Act, 1904, empowering the government to protect and maintain historical monuments and archaeological sites, and reorganised the Archaeological Survey of India under John Marshall.
- **Pair 3 is incorrectly matched:** The Partition of Bengal (1905) was carried out under Lord Curzon, not Lord Dufferin. Lord Dufferin (Viceroy 1884–1888) is instead associated with the Third Anglo-Burmese War and the annexation of Upper Burma (1885–86), as well as being Viceroy during the founding year of the Indian National Congress (1885), though he did not initiate it himself.

Q 133. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Gopabandhu Choudhury, a former Deputy Collector who resigned to join the freedom struggle, led the Salt Satyagraha at Inchudi, Balasore (coastal Odisha) in April 1930, marching with volunteers from Cuttack’s Swaraj Ashram. This movement was ranked second in significance only to Gandhi’s own Dandi Satyagraha, and became known as “the second Dandi.”
- **Pair 2 is correctly matched:** Jadunandan (Yadunandan) Sharma, second-in-command to Swami Sahajanand Saraswati, was the central peasant leader in Gaya district, Bihar, spearheading agitations like the Reora and Sandako Satyagrahas against zamindari exploitation and bakasht land seizures during the Kisan Sabha movement of the 1930s.
- **Pair 3 is correctly matched:** A.K. Gopalan served as the volunteer captain of the Guruvayoor Satyagraha (1931–32) in Kerala, working under K. Kelappan to demand temple entry for the depressed classes — he was notably assaulted by opponents of the movement during the campaign.
- **Pair 4 is correctly matched:** Maulana Abdul Hamid Khan Bhashani organised the historic “Praja Sammelan” (peasant conference) at Sirajganj in December 1932, a mass gathering of Bengali Muslim peasants that brought him nationwide recognition as a peasant leader, part of his broader mobilisation of migrant peasants across Assam and Bengal.

Q 134. Ans. B

Explanation:

- Rashbehari Bose was implicated in the 1912 Delhi-Lahore Conspiracy Case — the attempted bombing of Viceroy Lord Hardinge during his ceremonial

entry into Delhi — and subsequently fled India to evade arrest by British authorities.

- He reached Japan in 1915, where he lived in exile for decades, eventually becoming a naturalised Japanese citizen and marrying a Japanese woman, while continuing to organise anti-British revolutionary activity from abroad and building strong contacts within Japanese political and military circles.
- Following Japan's rapid conquests in Southeast Asia during WWII, Rashbehari Bose became the founding President of the Indian Independence League in 1942, an umbrella organisation of the Indian diaspora across East and Southeast Asia dedicated to India's liberation.

Q 135. Ans. A

Explanation:

- **End of the Second World War (1945)** – The war ended in August 1945, weakening Britain's economic and military position and accelerating the process of Indian independence.
- **C. Rajagopalachari's CR Formula and the Gandhi–Jinnah Talks (1944)** – The CR Formula was proposed in 1944 as a basis for resolving the Hindu–Muslim deadlock, followed by the Gandhi–Jinnah Talks in September 1944.
- **Formation of the Interim Government by the Congress (2 September 1946)** – Following the Cabinet Mission Plan, the Interim Government headed by Jawaharlal Nehru was formed.
- **Announcement by Clement Attlee regarding the end of British rule in India (20 February 1947)** – British Prime Minister Clement Attlee declared that power would be transferred to Indian hands by June 1948, paving the way for independence.

Q 136. Ans. D

Explanation:

- **Kakori Train Robbery (9 August 1925)** – Revolutionaries of the Hindustan Republican Association (HRA) looted government treasury from a train near Kakori to raise funds for revolutionary activities.
- **Saunders was assassinated in Lahore (17 December 1928)** – Members of the Hindustan Socialist Republican Association (HSRA), including Bhagat Singh, Rajguru, and Chandrashekhar Azad, killed British police officer J.P. Saunders to avenge the death of Lala Lajpat Rai.
- **Bomb was thrown in the Central Legislative Assembly (8 April 1929)** – Bhagat Singh and

Batukeshwar Dutt threw non-lethal bombs and distributed pamphlets in the Central Legislative Assembly in Delhi to protest repressive laws, deliberately surrendering afterward.

- **Bhagat Singh, Rajguru and Sukhdev were hanged (23 March 1931)** – The three revolutionaries were executed in Lahore Central Jail for their role in the Saunders assassination.

Q 137. Ans. A

Explanation:

- Koenraad Elst, a Belgian Indologist and historian, described the Khilafat Movement (1919–1924) as “a purely retrograde and reactionary movement” and argued that it was intrinsically anti-nationalist. He criticized the Indian National Congress, particularly Mahatma Gandhi, for supporting the movement, contending that it mixed religious issues with the nationalist struggle.
- The Khilafat Movement was launched by Indian Muslims to protest the dismemberment of the Ottoman Caliphate after the First World War. Gandhi supported the movement to foster Hindu–Muslim unity and strengthen the Non-Cooperation Movement.

Q 138. Ans. B

Explanation:

- The slogan “Down with Imperialism” was popularised by Bhagat Singh during the Indian freedom struggle. It became widely known after Bhagat Singh and Batukeshwar Dutt threw non-lethal bombs in the Central Legislative Assembly on 8 April 1929. While surrendering, they raised the slogans “Inquilab Zindabad” and “Down with Imperialism” to protest against British colonial rule and inspire revolutionary nationalism.
- **Additional Facts:**
 - ✓ Subhas Chandra Bose – Popularly associated with slogans such as “Jai Hind” and “Give me blood, and I will give you freedom.”
 - ✓ Bal Gangadhar Tilak – Famous for “Swaraj is my birthright, and I shall have it.”
 - ✓ Mahatma Gandhi – Associated with slogans such as “Do or Die” during the Quit India Movement.

Q 139. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Tiruchirappalli Proclamation (1801) was actually issued by the Maruthu Brothers during the Poligar Wars in the Carnatic region to call for unified regional resistance

against British direct tax collection. Velu Thampi, on the other hand, led the revolt in Travancore (1808–1809) and issued the Kundara Proclamation (1809) to mobilize the masses against the high-handedness of the British Subsidiary Alliance.

- **Statement 2 is correct:** The Sanyasi-Fakir Rebellion (1763–1800) broke out in Bengal primarily due to the severe economic distress caused by the Famine of 1770 and the British restrictions placed on the movement of nomadic pilgrims. It featured strong Hindu-Muslim unity and was ultimately suppressed by Warren Hastings. It forms the historical backdrop of Bankim Chandra Chatterjee's famous novel, Anandamath.
- **Statement 3 is correct:** The Paika Rebellion (1817) took place in Odisha. The Paikas were the traditional, hereditary landed militia. They were driven to armed revolt, attacking police stations and EIC treasuries, under the leadership of Bakshi Jagabandhu because the British implemented extortionist land revenue policies and resumed (confiscated) their traditional rent-free jagirs (lands). The rebellion was suppressed by 1818.

Q 140. Ans: A

Explanation:

- **Statement 1 is correct:** The movement, established in 1913 as the Pacific Coast Hindustan Association, had its headquarters in San Francisco. It published a weekly newspaper called Ghadar and strictly maintained a secular nationalist ideology, uniting Hindus, Muslims, and Sikhs.
- **Statement 2 is incorrect:** Lala Har Dayal was the ideological leader and guiding force of the movement, whereas Sohan Singh Bhakna served as its first President.
- **Statement 3 is correct:** Seeing World War I as an opportunity, the Ghadarites planned a massive armed uprising in Punjab and various military cantonments for February 21, 1915. However, the plot failed because a British spy, Kirpal Singh, infiltrated their ranks and leaked the plan.
- **Statement 4 is incorrect:** The Ghadar Movement completely rejected peaceful constitutional reforms and dominion status. As their name (which means "mutiny" or "revolt") and their motto ("Angrezi Raj ka Dushman") suggest, their sole aim was the complete overthrow of British rule through a violent armed rebellion, heavily inspired by the Revolt of 1857.
- **Additional Facts:**
 - ✓ **Defence of India Act, 1915:** Following the failed mutiny and mass arrests, the British

colonial government enacted this draconian law to quickly crush the revolutionary threat, allowing for preventive detention and speedy trials without the right to appeal.

- ✓ **Other Key Figures:** Kartar Singh Sarabha, a young revolutionary who was executed at just 19 years old, was a major inspiration for Bhagat Singh. Taraknath Das, another prominent figure, published the anti-British newspaper Free Hindustan in North America.

Q 141. Ans. C

Explanation:

- **Logic :** $(1st\ number \times IIrd\ number) \div 4 = IIIrd\ number$
- $(20, 16, 80) \rightarrow (20 \times 16) \div 4 = 80$
- $(22, 28, 154) \rightarrow (22 \times 28) \div 4 = 154$
- Only option C follows the same logic.
- $(18, 24, 108) \rightarrow (18 \times 24) \div 4 = 108$
- Hence, option C is correct.

Q 142. Ans. D

Explanation:

- $K - 1 \rightarrow J - 1 \rightarrow I - 1 \rightarrow H - 1 \rightarrow G$
- $H - 3 \rightarrow E - 3 \rightarrow B - 3 \rightarrow Y - 3 \rightarrow V$
- $N + 2 \rightarrow P + 2 \rightarrow R + 2 \rightarrow T + 2 \rightarrow V$
- Hence, option D is correct.

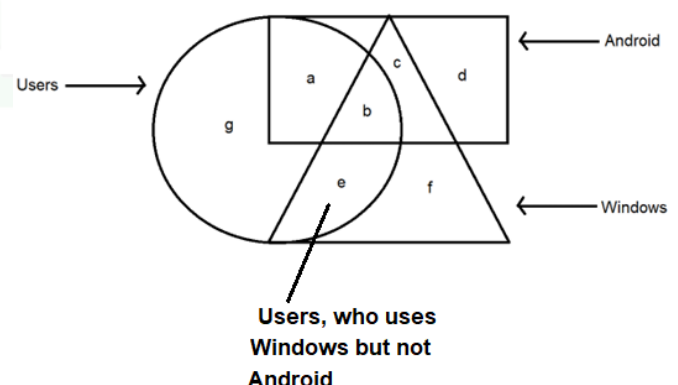
Q 143. Ans. D

Explanation:

- $442 - 17 + 8 \times 18 \div 25 = 145$
- **By using option D:**
- $442 \div 17 + 8 \times 18 - 25 = 145$
- $26 + 144 - 25 = 145$
- $145 = 145$
- Hence, option D is correct.

Q 144. Ans. C

Explanation:



From the above option, the alphabet “e” represents those users who uses windows but not android.

Hence, option C.

Q 145. Ans. B

Explanation:

- The statement talks about how electricity consumption will soon exceed overall energy demand by 2040 because air conditioners are coming into greater use. This rests on the assumption that air conditioners consume a lot of electricity. It follows. Thus, (b) is the right answer.
- I does not follow as the statement gives no information about electricity supply.

Q 146. Ans. B

Explanation:

- According to the question,
- 3 tonnes = 30 quintals
- Selling Price = $30 \times 250 = ₹7,500$
- Cost Price = ₹6,750
- Profit = $7,500 - 6,750 = ₹750$
- Profit Percentage = $750/6750 \times 100\% = 11(1/9)\%$

Q 147. Ans. C

Explanation:

- According to the question,
- ABCD is a cyclic quadrilateral.
- In a cyclic quadrilateral, the sum of opposite angles is 180° .
- Therefore,
- $A + C = 180^\circ$
- $(50^\circ + x) + (70^\circ + y) = 180^\circ$
- $x + y = 60^\circ$
- And,
- $B + D = 180^\circ$
- $(90^\circ - y) + 2x = 180^\circ$
- $2x - y = 90^\circ$
- Now,
- $x + y = 60^\circ$
- $2x - y = 90^\circ$
- Adding both equations,
- $3x = 150^\circ$
- $x = 50^\circ$
- Thus, $x = 50^\circ$ is obtained.

Q 148. Ans. B

Explanation:

- According to the question,
- The train stops at 3 stations.
- Duration of each stop = 20 minutes

- Total stopping time = $3 \times 20 = 60$ minutes = 1 hour
- Total time = 17 hours
- Therefore, actual travel time = $17 - 1 = 16$ hours
- Speed (S) = Distance / Time
- $= 1280 / 16$
- $= 80$ km/hr

Q 149. Ans. A

Explanation:

- According to the question,
- Principal (P) = ₹15,000
- Annual interest rate = 20%
- For half-yearly compounding,
- Interest rate per half-year = $20/2 = 10\%$
- Time = 1.5 years
- Therefore, total half-yearly periods = $1.5 \times 2 = 3$
- Amount (A) = $P \left(1 + \frac{r}{100}\right)^n$
- $A = 15000 \left(1 + \frac{10}{100}\right)^3$
- $A = 15000 \left(\frac{11}{10}\right)^3$
- $= 15000 \times 1.331$
- $= 19965$
- Compound Interest = $A - P$
- $= 19965 - 15000$
- $= ₹4,965$

Q 150. Ans. D

Explanation:

- According to the question,
- The number of students in the three classes is $3x$, $4x$, and $5x$, respectively.
- The new ratio after an increase of 10 students in each class is:
- $(3x+10) : (4x+10) : (5x+10) = 5 : 6 : 7$
- $(3x+10) : (4x+10) = 5 : 6$
- $6 \times (3x+10) = 5 \times (4x+10)$
- $18x + 60 = 20x + 50$
- $2x = 10$
- $x = 5$
- Total initial number of students = $3x + 4x + 5x = 12x$
- $= 12 \times 5$
- $= 60$

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